

Polyconvex neural-network surrogates with learned paired directional features for anisotropic hyperelastic homogenization

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Abstract

Approximating the effective anisotropic hyperelastic response of a representative volume element (RVE) requires constitutive expressiveness, mechanical consistency, and computational efficiency. Non-intrusive neural surrogates provide direct constitutive evaluations, whereas intrusive projection-based reduced-order models (PROMs) approximate the microscopic state and equilibrium equations. This work develops a polyconvex physics-augmented neural network (PANN) construction based on paired directional features with trainable orientations, stretch powers, and determinant exponents. The features are convex in independent deformation minors and are composed with convex, componentwise nondecreasing cores realized as an input-convex neural network (ICNN) or an input-convex Kolmogorov–Arnold network (ICKAN). The complete energies satisfy objectivity, an exactly stress-free reference, two-dimensional polyconvexity, and prescribed growth conditions by construction; global nonnegative energy is assessed through a separate sufficient bound for the trained parameters. On a periodic cell with four nonaligned cavities, learning six paired features lowers the held-out stress error sevenfold, to about 0.04%. On a second cell, with a rotated elliptical void, the selected polyconvex models attain approximately 0.63%. An unconstrained energy trained identically is more accurate but, beyond the data, loses rank-one convexity and remains finite under volume collapse. In a tensile-coupon FE^2 application against a full-order reference, the ICNN and ICKAN give displacement errors below 0.42%, and hyperreduced PROMs provide a complementary accuracy–cost benchmark. Under confined compression beyond the data, the polyconvex energies reach 500 MPa, whereas the unconstrained energy reaches less than half that load, where its rank-one curvature turns negative. These results quantify what the guarantees cost in accuracy and what they secure beyond the data.

Keywords: polyconvexity; physics-augmented neural networks; anisotropic hyperelasticity; computational homogenization; FE^2 ; input-convex neural networks; projection-based reduced-order models.

1 Introduction

The effective response of a heterogeneous material emerges from the deformation of its constituents and their interactions with the microstructure. At finite strains, this interaction can produce anisotropy and coupling between loading modes that are difficult to capture with a predetermined macroscopic constitutive law. Computational homogenization addresses this difficulty by resolving a representative volume element (RVE), in which an imposed deformation drives a microscopic boundary-value problem whose solution yields the effective stresses and constitutive derivatives [1–3]. The effective law is thus computed rather than prescribed in advance.

This flexibility comes at a computational cost. Characterizing the material over a range of deformations already requires many nonlinear finite element solves; using the same RVE within a structural calculation multiplies those solves across macroscopic integration points and load increments [3, 4]. Much of this repeated work concerns closely related microscopic states. Surrogate modeling seeks to exploit that relationship by computing representative solutions offline and reusing their information in less expensive online evaluations. The goal is to reduce the cost of these evaluations without losing the constitutive behavior that made the RVE description useful.

Two complementary approximation routes are relevant, but they play different roles in this work. Non-intrusive constitutive surrogates learn the effective response from RVE input–output data and dispense with

the microscopic solve online [5]. Intrusive projection-based reduced-order models (PROMs) instead compress the microscopic displacement field and retain projected equilibrium equations [6, 19, 20], with hyperreduction limiting the local contributions required to evaluate them [34]. This work follows the first route and learns the effective stored energy while embedding the constitutive restrictions needed for its mechanical use. The intrusive route is retained as a secondary reference because it preserves an approximate microscopic state and the underlying constituent law.

The central contribution is a family of learnable paired directional features for polyconvex anisotropic energies. Their orientations and exponents adapt to the RVE response, while their paired construction makes the stress vanish exactly at the undeformed state without compromising polyconvexity. Coupling these features to either an input-convex neural network (ICNN) or an input-convex Kolmogorov–Arnold network (ICKAN) gives two realizations of the same constitutive design. PROM-based surrogates of the same RVE then provide secondary accuracy–cost references. Two structural examples follow, a tensile coupon and a confined compression test beyond the data, to show how the learned energies behave once they leave isolated material evaluations.

1.1 Energy-based constitutive learning and polyconvexity

Direct constitutive learning moves the expensive microscopic solves to the data-generation stage and replaces them online by a trained material model. Neural homogenization has demonstrated this separation [7], making systematic sampling of representative microscopic responses an essential part of the offline workflow [8]. What the surrogate must represent depends on the material class. History-dependent responses may require learned history-to-response maps [9] or internal descriptors such as maximum directional stretches [10], while interacting scales may motivate staggered data generation and training [11]. The present work, by contrast, considers history-independent hyperelasticity. This removes memory from the state but requires the stress to derive from a stored-energy function.

Learning that function, rather than fitting each stress component independently, makes the stress work conservative and gives a major-symmetric constitutive tangent by differentiation. This is a central motivation for the mechanics-informed energy formulation of As’ad et al. [12]. Expressing the energy through objective deformation measures then removes dependence on a superposed rigid rotation. In this way, physics-augmented neural networks (PANNs) shift part of the burden from learning mechanical identities from data to satisfying them through the representation itself [13, 14]. Prediction accuracy remains to be established, but potential structure and objectivity need not depend on how densely the training domain has been sampled.

Preserving a potential does not, by itself, control its curvature. At finite strains, that control must also be imposed in appropriate variables. Convexity in Green–Lagrange strain [12] is attractive because stress and tangent follow directly from the learned energy. Its implication for curvature with respect to the deformation gradient nevertheless contains a stress-dependent term [15], so it is not a global polyconvexity guarantee. Conversely, Xu et al. [16] parameterize an incremental tangent through a lower-triangular factorization to enforce positive semidefiniteness. This controls the tangent matrix but does not define the scalar potential required for integrability. The modeling task is therefore to preserve both the potential and the desired finite-strain curvature condition within one construction.

Polyconvexity provides a route to that combination. It represents the energy as a convex function of deformation minors, yielding a sufficient condition for rank-one convexity [17]. For the two-dimensional model considered here, the independent arguments are the deformation gradient and its determinant. This framework also underpins existence results when the additional coercivity and admissibility hypotheses are met [18]. Polyconvexity serves here as a sufficient constitutive restriction, not as a necessary property of every homogenized material or a guarantee of structural uniqueness. The remaining challenge is to keep enough freedom within these restrictions to describe the anisotropic RVE response.

1.2 Directional features and representational expressiveness

For anisotropy, this challenge is especially consequential, because a flexible scalar core can distinguish only the deformation states that its inputs distinguish. Structural tensors encode material directions in objective scalar quantities and underpin established anisotropic polyconvex energies [40]. The directional description is itself a constitutive choice. Models of fiber-reinforced tissue, for example, must account for a distribution of orientations rather than one perfectly aligned family when that dispersion affects the mechanical response [41]. A learned feature map must likewise expose the relevant directional distinctions while remaining compatible with objectivity and convexity in the deformation minors. It is therefore part of the constitutive model, not merely a preprocessing step.

A convex scalar core does not resolve this choice. An ICNN provides a convex function [42], but a sufficient composition with nonlinear features requires those features to be convex in the independent deformation

minors and the core to be componentwise nondecreasing [13]. This monotonicity imposes an order on feature space, since $\mathbf{z}(\mathbf{F}_a) \leq \mathbf{z}(\mathbf{F}_b)$ componentwise implies $H(\mathbf{z}(\mathbf{F}_a)) \leq H(\mathbf{z}(\mathbf{F}_b))$. Increasing the network width cannot reverse that ordering. Reference corrections and growth terms, added later to complete the energy, do not remove this restriction on the learned contribution. A poorly chosen feature family can therefore impose a representational obstruction before network capacity or optimization becomes limiting.

The numerical evidence of Klein et al. [13] illustrates this distinction. For their cubic beam-lattice data, an exactly objective invariant-based polyconvex model fails to represent the shear response, whereas a model formulated in $\mathbf{F}$, $\text{cof } \mathbf{F}$, and $\det \mathbf{F}$ gives a substantially better fit while approximating objectivity through data augmentation. The same invariant-based model performs very well for their analytically generated transversely isotropic data. In these results, accuracy depends on whether the admissible inputs expose the distinctions required by the material response, rather than on polyconvexity alone. This motivates enlarging the feature family before concluding that a convex core is inadequate.

The natural response is to make the directional description adaptive without abandoning the admissible composition. Generalized-structure-tensor models calibrate the structural tensors together with the energy model, allowing the data to identify aspects of the anisotropy [43]. Recent triclinic invariant constructions and group symmetrization further broaden the polyconvex descriptions available for finite material-symmetry groups [44]. Greater directional freedom, however, must be reconciled with the undeformed state. A learned anisotropic contribution need not produce zero reference stress, while cancelling that stress with an arbitrary strain correction can violate polyconvexity or material symmetry. The invariant-based normalization and analytic growth terms of Linden et al. [14] demonstrate why these operations must belong to the admissible constitutive construction. The feature family must therefore do two jobs at once: distinguish the relevant directional response and control its reference derivatives so that exact stress normalization remains compatible with polyconvexity. This is the role of the paired features introduced here.

While the feature map determines what the model can distinguish, the scalar core determines how those distinctions are combined into an energy. Prior constructions distribute this flexibility in different ways. Constitutive artificial neural networks (CANNs) learn coefficients and exponents within an interpretable catalog of admissible energy terms [45], whereas neural ordinary differential equations can learn monotone energy derivatives and recover convex energy contributions by integration [46]. In the present work, an ICNN combines the paired features through nonnegative weights and fixed convex activations. A Kolmogorov–Arnold network (KAN) instead places trainable univariate functions on its edges, a flexibility explored in constitutive KANs [47]. The feature-composition argument holds only if these edge functions are convex and nondecreasing, as in monotone input-convex KANs (ICKANs) [48]. A smooth ICNN and an integrated-hat ICKAN are compared under the same paired features and analytic completion. The comparison isolates how the scalar dependence is represented, while the directional information and the mechanical guarantees stay the same.

1.3 Projection-based reduced-order models

The preceding approach learns the effective energy and thereby removes the microscopic solve from the online stage. Projection-based reduction makes a different compromise, retaining the RVE boundary-value problem but seeking its solution in a space learned from previous microscopic states. In an affine PROM, those states define a basis, commonly obtained through proper orthogonal decomposition (POD), and projected equilibrium determines the coordinates of each new displacement field. Reduced finite-strain homogenization has shown that this strategy can reuse information from earlier RVE solves without replacing the constituent law [19–21]. PROMs therefore provide a natural physics-based reference for the present study, since they preserve more of the microscopic problem than an effective-energy surrogate at the price of an online equilibrium solve.

The affordability of that solve depends first on the geometry of the solution family. A small number of loading parameters does not necessarily imply that the corresponding microscopic states lie close to a small global linear space. When their Kolmogorov n -width decays slowly, many basis vectors, and hence many equilibrium unknowns, are required to achieve adequate accuracy. Considerable effort has been devoted to mitigating this Kolmogorov barrier with richer reduced representations, from local linear spaces [25] to nonlinear manifolds built with autoencoders [26], quadratic corrections [27], or learned closures in a POD coordinate space [22–24, 50]. The last of these, PROM-ANN [22], offers the compromise sought here, because it enriches the geometry of the reduced state without giving up either its POD description or projected equilibrium. Equilibrium determines only a small set of primary coordinates, while a regression model, here an artificial neural network (ANN), reconstructs the secondary coordinates that populate a much richer POD expansion [24]. The microscopic displacement can thereby draw on many modes while the online nonlinear system remains small, which motivates the primary–secondary construction adopted for the RVE in this work.

Reducing the number of unknowns does not make the reduced equations inexpensive, because assembling their residual and tangent may still require constitutive evaluations over the complete microscopic mesh at every iteration. While hardware acceleration performs this full-mesh work faster [28], hyperreduction reduces how much of it must be performed, turning a PROM into a hyperreduced PROM (HPROM). Methods that act directly on the projected equations [29, 30], instead of first reconstructing high-dimensional quantities as discrete empirical interpolation does [31], include the linear-programming empirical quadrature procedure [35, 36], energy-conserving sampling and weighting (ECSW) [32, 33], and the empirical cubature method (ECM) [21, 34]. Each replaces full assembly by a nonnegative weighted sum of selected element contributions while leaving the local constituent evaluations intact. This work follows ECM because previous developments extend its fixed rule, in which one support and one weight vector serve the whole training space, toward movable points [37], subspace-adaptive weights [38], and finally the manifold-adaptive weights ECM (MAW-ECM) [39], which is the path needed to couple hyperreduction with PROM-ANN. In MAW-ECM, the weights become fields over the latent coordinates, so the same primary coordinates that locate the microscopic state on the PROM-ANN manifold also select the rule used to evaluate its equilibrium and stress.

Together, these choices yield three intrusive references for the structural comparison. The affine HPROM retains a linear POD state, solves projected equilibrium, and evaluates it with fixed ECM. HPROM-ANN replaces the linear state by the primary-secondary manifold and the fixed rule by MAW-ECM, but still determines its primary coordinates through equilibrium. Its direct variant, D-HPROM-ANN, removes that final online solve by evaluating the decoder at the imposed strain, as in the hyperelastic example of [39]. Moving across the three models removes successive portions of the online microscopic work while preserving an approximate microscopic state and selected evaluations of the constituent law. They contextualize the accuracy and cost of the proposed energy surrogates but do not share their constitutive guarantees, because the approximate effective stress of a reduced model need not derive from a single effective potential, even when the constituent law is retained and the cubature weights are nonnegative.

1.4 Contribution and outline

Against these intrusive physics-based references, the central route pursued in this work is an effective-energy surrogate whose anisotropy is learned without relinquishing polyconvexity or exact reference-state normalization. The construction rests on paired directional features with trainable orientations, stretch powers, and determinant exponents constrained by explicit joint-convexity bounds.

The pairing is designed so that the strain derivative of every feature at the undeformed state is proportional to the identity tensor, even though its response away from that state remains anisotropic. The complete learned contribution can consequently be normalized by a single scalar correction affine in the determinant. This correction cancels the reference stress without tying that cancellation to the independently adjustable volumetric stiffness, allowing the directional description to adapt to the RVE while the reference state remains under analytic control.

With the directional information in this form, two monotone convex cores provide alternative representations of its scalar energetic dependence: a smooth ICNN and an integrated-hat spline ICKAN. In both cases, the learned core, reference correction, and growth terms form one complete energy for which the analysis establishes objectivity, an exactly stress-free reference, two-dimensional polyconvexity, and the specified growth limits. Global nonnegative energy is treated separately through a sufficient lower bound evaluated for the trained parameters. The novelty lies in bringing established convex-network and affine-determinant normalization principles together through the paired anisotropic family and its explicit analytical conditions.

The assessment works at two levels. At the constitutive level, comparing the constrained energies on independent RVE responses with the Unconstrained energy, a network trained under the same protocol without the curvature and growth constraints, measures the accuracy that the guarantees cost. The same study compares learned with fixed paired directions and locates where the Unconstrained energy loses rank-one convexity beyond the data, while checks of the tangent and of the sign of the energy probe behavior that a fit metric cannot reveal [49]. At the structural level, the learned energies replace the RVE query of the first-order two-level finite element formulation, denoted FE^2 , in which an RVE problem supplies the material response at each macroscopic integration point [4]. In a tensile coupon, they are compared with a nested full-order model (FOM)- FE^2 reference, and the three reduced models described above serve as physics-based accuracy-cost benchmarks. A confined compression test then drives the same energies beyond the data to ask which of them keep the structural problem solvable.

The conclusions are tied to the two periodic cells, plane-strain kinematics, and the two structural loadings. Agreement with FOM- FE^2 measures surrogate error within the adopted homogenization model, not error relative to a structure in which every microscopic cell is resolved explicitly, and the reported timings cover the online workflows only.

The remainder of the paper is organized as follows. Section 2 defines the homogenization problem and work-conjugate variables. Section 3 introduces the admissibility criteria, constructs the learned energies, and proves their properties, after which Section 4 defines the reduced microscopic references used in the structural comparison. Section 5 reports the numerical examples, first the multicavity RVE (MC-RVE) representation study and then the single-cavity RVE (SC-RVE) deployment study, which ends with the coupon FE² comparison and a confined compression test beyond the data. Section 6 closes the paper, and the supplementary material documents additional curvature and closed-cycle diagnostics.

2 Finite-strain homogenization and the reference problem

The two surrogate routes replace different parts of the online calculation, but both approximate the same constitutive map, which returns the effective response of a periodic RVE for a prescribed macroscopic deformation. The FOM of the RVE supplies the reference by solving microscopic equilibrium. This section establishes the kinematic input, periodic cell problem, effective outputs, and stress–tangent interface shared by all models before their approximation strategies diverge. The cell geometry and material parameters are specified in the numerical examples.

2.1 Kinematics and work-conjugate variables

Let $\mathbf{F} \in \text{GL}^+(2)$ denote the prescribed in-plane macroscopic deformation gradient. Its associated right Cauchy–Green tensor and Green–Lagrange strain are

$$\mathbf{C} = \mathbf{F}^T \mathbf{F}, \quad \mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I}), \quad J = \det \mathbf{F} > 0. \quad (1)$$

The macroscopic input is written simply as $\mathbf{F}$ to match the constitutive notation used throughout the paper. The subscript μ marks microscopic quantities, whereas the unsubscripted W , $\mathbf{P}$, and $\mathbf{S}$ denote effective quantities obtained by homogenizing the cell. Plane strain is adopted, so the corresponding three-dimensional gradient is $\text{diag}(\mathbf{F}, 1)$, and the same embedding applies locally to $\mathbf{F}_\mu$. This constrains the out-of-plane deformation, not the out-of-plane stress.

The constitutive input used throughout is the three-component engineering representation of the Green–Lagrange strain. With $\mathbf{S}$ the effective second Piola–Kirchhoff stress, the work-conjugate convention is

$$\mathbf{e} = (E_{11}, E_{22}, 2E_{12})^T, \quad \mathbf{s} = (S_{11}, S_{22}, S_{12})^T, \quad \mathbf{S} : d\mathbf{E} = \mathbf{s} \cdot d\mathbf{e}. \quad (2)$$

Thus $\gamma_{12} = 2E_{12}$ is paired with S_{12} , with no additional factor of two in the stress vector. For a differentiable effective energy density $W(\mathbf{e})$, this convention gives the stress and the constitutive tangent

$$\mathbf{s} = \frac{\partial W}{\partial \mathbf{e}}, \quad \mathbf{D} = \frac{\partial \mathbf{s}}{\partial \mathbf{e}} = \frac{\partial^2 W}{\partial \mathbf{e} \partial \mathbf{e}} \in \mathbb{R}^{3 \times 3}, \quad D_{ij} = \frac{\partial s_i}{\partial e_j}. \quad (3)$$

This convention is used consistently in training, work integrals, and tangent checks.

An admissible strain input must satisfy $\mathbf{I} + 2\mathbf{E} \succ 0$. The RVE is queried with the representative

$$\mathbf{F}(\mathbf{e}) = \sqrt{\mathbf{I} + 2\mathbf{E}}, \quad (4)$$

where the square root is symmetric positive definite. Objectivity allows a superposed macroscopic rotation to be removed without changing the energy or second Piola–Kirchhoff stress. The choice remains compatible with anisotropy, because material directions stay fixed in the reference configuration, and it places no symmetry requirement on the local microscopic gradients.

2.2 Periodic microscopic equilibrium

Let Y denote the reference periodic cell and $Y_s \subset Y$ its solid part. The prescribed $\mathbf{F}$ determines the affine part of the microscopic deformation, whereas the fluctuation $\mathbf{w}$ is unknown,

$$\boldsymbol{\varphi}(\mathbf{X}) = \mathbf{F}\mathbf{X} + \mathbf{w}(\mathbf{X}), \quad \mathbf{F}_\mu = \mathbf{F} + \nabla \mathbf{w}, \quad (5)$$

with gradients taken in the reference configuration. Periodicity requires $\mathbf{w}(\mathbf{X}^+) = \mathbf{w}(\mathbf{X}^-)$ at paired points on opposite cell faces. Consequently, the microscopic displacement $\mathbf{u}_\mu = \boldsymbol{\varphi} - \mathbf{X}$ satisfies

$$\mathbf{u}_\mu(\mathbf{X}^+) - \mathbf{u}_\mu(\mathbf{X}^-) = (\mathbf{F} - \mathbf{I})(\mathbf{X}^+ - \mathbf{X}^-). \quad (6)$$

The finite-strain displacement jump therefore depends on $\mathbf{F} - \mathbf{I}$, not directly on $\mathbf{E}$. Fixing the fluctuation at one node removes the arbitrary translation without clamping the cell boundary. The internal void surface is traction-free.

Let $\psi_\mu(\mathbf{X}, \mathbf{F}_\mu)$ be the solid's objective hyperelastic energy per unit reference volume and $\mathbf{P}_\mu = \partial_{\mathbf{F}_\mu} \psi_\mu$ its first Piola–Kirchhoff stress. In the absence of microscopic body forces, equilibrium requires

$$\int_{Y_s} \mathbf{P}_\mu : \nabla \delta \mathbf{w} \, dV = 0 \quad (7)$$

for every admissible periodic test fluctuation. Periodic kinematics and the associated traction balance provide the standard work-consistent cell problem [1]. After finite element discretization and elimination of the periodic constraints, let $\mathbf{a}$ collect the independent fluctuation coordinates. The full-order equilibrium problem is

$$\mathbf{r}(\mathbf{a}, \mathbf{e}) = \partial_{\mathbf{a}} \Pi_\mu(\mathbf{a}, \mathbf{e}) = \mathbf{0}, \quad \Pi_\mu = \int_{Y_s} \psi_\mu(\mathbf{X}, \mathbf{F}_\mu) \, dV, \quad (8)$$

where the strain dependence enters through Eq. (4). Solving Eq. (8) gives $\mathbf{a}^*(\mathbf{e})$ and hence the microscopic fields. The microscopic deformation must also remain orientation-preserving, $\det \mathbf{F}_\mu > 0$.

2.3 Effective energy and stress

On the selected equilibrium branch, the effective energy and stresses are

$$W(\mathbf{e}) = \frac{\Pi_\mu(\mathbf{a}^*(\mathbf{e}), \mathbf{e})}{|Y|}, \quad \mathbf{P} = \frac{1}{|Y|} \int_{Y_s} \mathbf{P}_\mu \, dV, \quad \mathbf{S} = \mathbf{F}^{-1} \mathbf{P}. \quad (9)$$

The normalizing volume $|Y|$ includes the void, so these quantities describe the porous cell as a whole. For the two-dimensional discretization, the same reference thickness enters the integration weights and the cell volume, and cancels in their ratio. The stress conversion in Eq. (9) is performed *after* averaging the first Piola–Kirchhoff stress. In general, averaging microscopic second Piola–Kirchhoff stresses does not produce $\mathbf{S}$, because $\mathbf{F}_\mu$ varies within the cell.

The connection between energy and averaged stress follows from equilibrium. For a macroscopic variation, $\delta \mathbf{F}_\mu = \delta \mathbf{F} + \nabla \delta \mathbf{w}$, and Eq. (7) removes the fluctuation contribution, which yields

$$\frac{1}{|Y|} \int_{Y_s} \mathbf{P}_\mu : \delta \mathbf{F}_\mu \, dV = \mathbf{P} : \delta \mathbf{F} = \mathbf{S} : \delta \mathbf{E} = \mathbf{s} \cdot \delta \mathbf{e}. \quad (10)$$

This is the Hill–Mandel work-consistency condition [1], which also establishes $\mathbf{s} = \partial_{\mathbf{e}} W$ on a smooth branch.

The Newton solver follows an equilibrium branch, possibly through load continuation, and its convergence establishes stationarity rather than global minimization of Π_μ or uniqueness of the cell response. The resulting smooth branch serves as the reference constitutive response over the investigated states, without assuming that microscopic polyconvexity makes the homogenized energy polyconvex.

2.4 Consistent constitutive tangent

The tangent must account for the change in microscopic equilibrium as the input strain changes. To make this dependence explicit, write the stress evaluator before substitution of the equilibrium state as $\widehat{\mathbf{s}}(\mathbf{a}, \mathbf{e})$, so that $\mathbf{s}(\mathbf{e}) = \widehat{\mathbf{s}}(\mathbf{a}^*(\mathbf{e}), \mathbf{e})$. If $\mathbf{r}_{,\mathbf{a}}$ is nonsingular on the selected branch, implicit differentiation of Eq. (8) gives

$$\frac{d\mathbf{a}^*}{d\mathbf{e}} = -\mathbf{r}_{,\mathbf{a}}^{-1} \mathbf{r}_{,\mathbf{e}}, \quad \mathbf{D} = \widehat{\mathbf{s}}_{,\mathbf{e}} - \widehat{\mathbf{s}}_{,\mathbf{a}} \mathbf{r}_{,\mathbf{a}}^{-1} \mathbf{r}_{,\mathbf{e}}. \quad (11)$$

All partial derivatives are evaluated at equilibrium, holding the other argument fixed. The first term differentiates the stress at fixed fluctuation coordinates, and the second accounts for microscopic relaxation. Omitting that correction generally does not give the derivative of the deployed constitutive response. For the smooth hyperelastic branch, the result is also $\partial_{\mathbf{e}}^2 W$.

No explicit matrix inverse is needed in Eq. (11), because the sensitivities follow from linear systems with the converged microscopic tangent. The reference implementation uses one nonlinear equilibrium solution followed by three sensitivity right-hand sides, one per strain component. Independent finite-difference checks, which perturb the input and re-solve equilibrium, verify this linearization. The same consistency requirement applies to each surrogate, whose tangent must differentiate its own stress evaluation.

2.5 Macroscopic equilibrium and the constitutive interface

The pair $(\mathbf{s}, \mathbf{D})$ is the common interface through which every constitutive evaluator enters the structural problem. Let $\mathbf{U}$ collect the free macroscopic nodal displacements, and let $\mathbf{e}_g(\mathbf{U})$ be the engineering Green strain at integration point g . With $\mathbf{B}_g = \partial \mathbf{e}_g / \partial \mathbf{U}$, reference-volume integration weights Ω_g , and dead external loads $\mathbf{f}_{\text{ext}}$, equilibrium reads

$$\mathbf{R}_M(\mathbf{U}) = \sum_g \Omega_g \mathbf{B}_g^T \mathbf{s}_g(\mathbf{e}_g(\mathbf{U})) - \mathbf{f}_{\text{ext}} = \mathbf{0}. \quad (12)$$

Its tangent contains material and geometric contributions,

$$\mathbf{K}_M = \sum_g \Omega_g \left[\mathbf{B}_g^T \mathbf{D}_g \mathbf{B}_g + \sum_{\alpha=1}^3 (s_g)_\alpha \frac{\partial^2 (e_g)_\alpha}{\partial \mathbf{U} \partial \mathbf{U}} \right]. \quad (13)$$

The second term arises because the Green strain is nonlinear in the displacements. A structural Newton solver therefore needs both $\mathbf{s}_g$ and $\mathbf{D}_g$, irrespective of how the constitutive response is computed.

The nested FOM-FE² evaluation provides the reference instance of this interface, in which every constitutive query solves the full periodic RVE problem defined above. The learned energies of Section 3 return $\mathbf{s}$ and $\mathbf{D}$ by differentiation and remove the microscopic solve from the online stage. The reduced models of Section 4 instead keep an approximate microscopic state. The affine HPROM and HPROM-ANN obtain it by solving equilibrium in reduced coordinates with hyperreduced integration, whereas D-HPROM-ANN evaluates it without that solve. Despite these different routes, all models supply the same interface while the macroscopic mesh, loading, and equilibrium algorithm remain unchanged.

Holding that structural problem fixed makes the nested FOM-FE² solution the reference for measuring surrogate errors at fixed macro and micro discretizations, rather than a direct numerical simulation resolving every pore in the coupon. Homogenization error, boundary-layer effects, and mesh-discretization error are therefore distinct from the surrogate errors reported below. The cell itself is an ideal periodic material model, without a claim of statistical representativeness for a random microstructure.

3 Learnable paired-feature polyconvex energies

The non-intrusive route developed here replaces the repeated online solution of the microscopic problem by a learned effective stored energy. Its central constitutive relation is

$$\mathbf{s}(\mathbf{e}) = \nabla_{\mathbf{e}} W(\mathbf{e}), \quad (14)$$

where $\mathbf{s}$ is the stress work-conjugate to the engineering-strain vector $\mathbf{e}$. All stress components then derive from a common potential, and the Hessian of W supplies a major-symmetric constitutive tangent.

One could stop at this point and construct a flexible hyperelastic network without restricting the curvature of its learned potential. With objective inputs and a stress-free reference state, this option defines the *Unconstrained energy* (Section 3.2), which shows what can be represented before stronger finite-strain restrictions are introduced.

Potential structure and reference normalization alone do not control deformation-gradient curvature or prevent finite energy as $J \rightarrow 0^+$. This work instead pursues polyconvexity by constructing the energy as a convex function of independent deformation minors, together with analytic terms that control the reference state and growth. The learned contribution combines a common family of paired directional features with one of two monotone convex cores, an ICNN or an ICKAN. These cores provide alternative representations while preserving the same feature construction and mechanical guarantees.

Figure 1 summarizes the construction, tracing the path from RVE data through the paired directional features and either convex core to the analytic completion of the stored energy, whose derivatives supply stress and tangent. Throughout, $\mathbf{F} \in \text{GL}^+(2)$, so that $J = \det \mathbf{F} > 0$, and derivatives with respect to deformation are taken with the learned parameters fixed.

3.1 Admissibility criteria and scope of the guarantees

A small prediction error is not sufficient to define an admissible hyperelastic surrogate. Six requirements, standard in constitutive PANN formulations [13, 14] and labeled here for later reference, organize the construction:

(C1) *Potential structure*: stress is the derivative of a single-valued stored energy in work-conjugate variables.

- (C2) *Objectivity*: a superposed rigid rotation does not change that energy.
- (C3) *Stress symmetry and anisotropy*: the second Piola–Kirchhoff stress is symmetric, while directional dependence is allowed without prescribing a material symmetry group.
- (C4) *Reference normalization*: the undeformed state has zero energy and zero stress.
- (C5) *Polyconvexity*: the energy admits a convex representation in deformation minors, which is sufficient for rank-one convexity.
- (C6) *Growth*: energy diverges under volume collapse, unbounded volume expansion, and unbounded distortion at bounded positive volume.

The Unconstrained energy satisfies (C1)–(C4) by construction, whereas (C5) and (C6) are not imposed on it. The ICNN and ICKAN energies satisfy all six criteria by construction for every finite parameter set that obeys the stated architectural constraints.¹ A property counts here as guaranteed only if it follows from the representation itself or from a sufficient bound under stated assumptions. Sampled numerical checks verify the implementation but do not replace such an argument.

Potential structure (C1), supplied by Eq. (14), implies that the work around every closed strain path Γ satisfies

$$\oint_{\Gamma} \mathbf{s} \cdot d\mathbf{e} = 0. \quad (15)$$

For the isothermal purely elastic setting considered here, (C1) likewise gives $\mathbf{s} \cdot \dot{\mathbf{e}} - \dot{W} = 0$, so the mechanical dissipation vanishes. The stress Jacobian is then the Hessian of W and obeys the cross-derivative identities $\partial s_i / \partial e_j = \partial s_j / \partial e_i$, that is, the major symmetry $\mathbf{D} = \mathbf{D}^T$ of the constitutive tangent in Eq. (3).

Objectivity (C2) must remove dependence on the spatial observer without inadvertently imposing material isotropy. For a superposed proper rotation $\mathbf{Q}$, it requires

$$W(\mathbf{Q}\mathbf{F}) = W(\mathbf{F}), \quad \mathbf{P}(\mathbf{Q}\mathbf{F}) = \mathbf{Q}\mathbf{P}(\mathbf{F}). \quad (16)$$

Using $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ as an energy input enforces the first identity, because a superposed spatial rotation leaves $\mathbf{C}$ unchanged. Differentiating the objective energy then gives the stress transformation. The stress symmetry required by (C3), $\mathbf{S} = \mathbf{S}^T$, follows for the energy models considered here from differentiation with respect to symmetric strain. It implies a symmetric Cauchy stress $\boldsymbol{\sigma} = J^{-1} \mathbf{F} \mathbf{S} \mathbf{F}^T$, as required by angular-momentum balance in the present continuum.

Objectivity still permits anisotropy, because material symmetry is a different invariance. A right action $\mathbf{F} \mapsto \mathbf{F}\mathbf{Q}_m$ leaves the energy unchanged precisely when $\mathbf{Q}_m$ belongs to the material symmetry group. Section 3.3 implements this distinction with directions attached to the reference material. A superposed spatial rotation acts on $\mathbf{F}$ from the left and does not alter the directional stretch measured in that reference. The major symmetry of $\mathbf{D}$ is a third, separate property, which follows from (C1) rather than from the symmetry of the stress tensor.

Reference normalization (C4) makes the learned energy consistent with the undeformed state by requiring

$$W(\mathbf{I}) = 0, \quad \mathbf{s}(\mathbf{0}) = \mathbf{0}, \quad (17)$$

where $\mathbf{s}(\mathbf{0}) = \mathbf{0}$ is equivalent to $\mathbf{P}(\mathbf{I}) = \mathbf{0}$. The first condition fixes an arbitrary additive energy constant. The second is a substantive stress condition, and the admissible correction depends on the model class. For the Unconstrained energy, Section 3.2 subtracts the raw reference value and a term linear in $\mathbf{E}$. That correction preserves its scalar potential, but an arbitrary linear function of $\mathbf{E}$ need not preserve polyconvexity in $\mathbf{F}$. The ICNN and ICKAN energies instead use a correction that is affine in the determinant, which preserves convexity in the independent minors (Section 3.5).

Polyconvexity (C5) supplies the curvature control that potential structure leaves open. Direct convexity in $\mathbf{F}$ is generally too restrictive for objective finite-strain energies with singular-volume growth, so convexity is imposed instead in independent deformation minors [17, 18]. In two dimensions,² this means that the energy admits the representation

$$W(\mathbf{F}) = \mathcal{P}(\mathbf{F}, \det \mathbf{F}), \quad \mathcal{P} \text{ convex in its independent arguments.} \quad (18)$$

¹Global nonnegative energy is not part of (C1)–(C6). It is checked separately for the trained parameters in Section 5.3.3, using the sufficient bound of Appendix C.

²The standard three-dimensional definition uses $(\mathbf{F}, \text{cof } \mathbf{F}, \det \mathbf{F})$. The two-dimensional guarantee established here does not imply an admissible extension to arbitrary out-of-plane deformation.

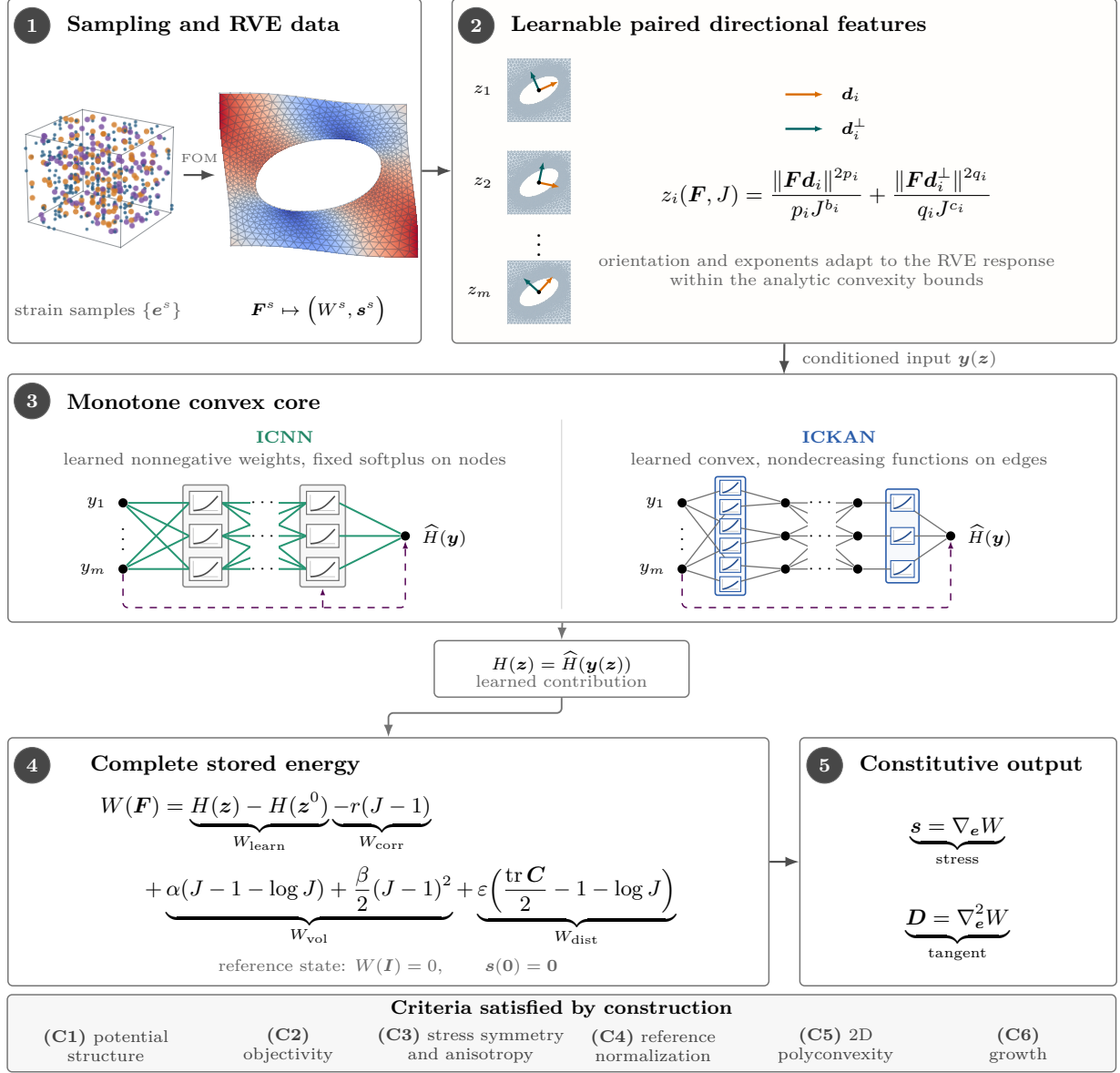


Figure 1: Overview of the proposed constitutive construction. Sampled strain states are evaluated with the periodic RVE model to supply energy and stress data. These responses train learnable paired directional features and a monotone convex ICNN or ICKAN core. Analytic reference and growth terms complete the stored energy. Differentiation then supplies the work-conjugate stress and constitutive tangent. Criteria (C1)–(C6), defined in Section 3.1, follow from the complete construction rather than from sampled numerical checks.

Here $\mathcal{P}$ is defined on the convex set $\mathbb{R}^{2 \times 2} \times (0, \infty)$, treating its matrix and scalar arguments as independent. Only its evaluation in W sets the second argument equal to $\det \mathbf{F}$. The cofactor is linear in the entries of a 2×2 matrix, so it need not be introduced as an additional independent argument. Convexity of $\mathcal{P}$ implies that W cannot curve downward along an admissible rank-one perturbation of $\mathbf{F}$.

Convexity in $\mathbf{C}$ or $\mathbf{E}$, as imposed in strain-space formulations such as that of As’ad et al. [12], does not imply polyconvexity [15], because $\mathbf{E} = \frac{1}{2}(\mathbf{F}^T \mathbf{F} - \mathbf{I})$ depends quadratically on $\mathbf{F}$. Appendix A derives the rank-one consequence of (C5), states its differential form, and gives an analytical counterexample for convexity in $\mathbf{E}$.

Growth (C6) guards against volume collapse and unbounded deformation, which a polyconvex energy may still allow. It requires $W \rightarrow \infty$ as $J \rightarrow 0^+$, as $J \rightarrow \infty$, and as $\|\mathbf{F}\| \rightarrow \infty$ while J remains bounded away from zero and infinity. These conditions contribute to coercivity. Existence of a minimizer also requires admissible boundary data and finite-energy competitors [18], and no constitutive barrier guarantees convergence of an unconstrained Newton iteration.

3.2 Unconstrained energy reference

The Unconstrained energy is a smooth neural network $\widehat{W}_\theta$ acting on

$$\mathbf{x}(\mathbf{e}) = (C_{11} - 1, C_{22} - 1, C_{12}, J - 1)^T. \quad (19)$$

Here $\mathbf{C} = \mathbf{I} + 2\mathbf{E}$ and $J = \sqrt{\det \mathbf{C}}$, so the four inputs are functions of the same three engineering-strain components. Using components of $\mathbf{C}$ in a material reference basis ensures objectivity under superposed spatial rotations but does not impose isotropy, since the network may depend differently on C_{11} , C_{22} , and C_{12} . Denote the resulting raw energy by $\widetilde{W}(\mathbf{e}) = \widehat{W}_\theta(\mathbf{x}(\mathbf{e}))$. A generic network does not satisfy the reference conditions automatically, so the normalized energy is defined, following the consistency correction of As’ad et al. [12], as

$$W_{\text{unc}}(\mathbf{e}) = \widetilde{W}(\mathbf{e}) - \widetilde{W}(\mathbf{0}) - \partial_{\mathbf{e}} \widetilde{W}(\mathbf{0}) \cdot \mathbf{e}. \quad (20)$$

By construction, $W_{\text{unc}}(\mathbf{0}) = 0$ and $\partial_{\mathbf{e}} W_{\text{unc}}(\mathbf{0}) = \mathbf{0}$. Because the correction is affine in $\mathbf{e}$, it leaves the Hessian of the raw network unchanged, so the reference energy and stress are fixed while the learned curvature remains unrestricted.

3.3 Feature design: expressiveness and admissibility

The first design question is how to describe deformation without erasing the effective anisotropy of the RVE. Classical invariants provide the natural starting point. In two dimensions, the principal invariants of $\mathbf{C}$ are $I_1 = \text{tr } \mathbf{C}$ and $\det \mathbf{C} = J^2$, and they determine all three invariants of the plane-strain embedding $\text{diag}(\mathbf{C}, 1)$. They describe principal stretches without recording their orientation relative to the material, so an energy depending only on them is isotropic.

To retain directional information, one can supplement the isotropic measures with a structural direction $\mathbf{d}$. The familiar anisotropic stretch invariant is

$$I_4(\mathbf{d}) = \mathbf{d}^T \mathbf{C} \mathbf{d} = \mathbf{C} : (\mathbf{d} \otimes \mathbf{d}) = \|\mathbf{F} \mathbf{d}\|^2, \quad \|\mathbf{d}\| = 1. \quad (21)$$

It measures squared stretch along a material direction and remains unchanged by a superposed spatial rotation. Such directional invariants are standard ingredients of anisotropic hyperelastic energies [40, 41]. Here $\mathbf{d}$ need not represent a physical fiber. Learned directions serve as objective scalar descriptors of the RVE’s effective anisotropy. In two dimensions, $\mathbf{d}$ is naturally accompanied by its perpendicular $\mathbf{d}^\perp$. For this orthonormal pair,

$$I_4(\mathbf{d}) + I_4(\mathbf{d}^\perp) = I_1, \quad \|\text{cof } \mathbf{F} \mathbf{d}\|^2 = I_4(\mathbf{d}^\perp). \quad (22)$$

The first identity shows that the pair splits I_1 between two perpendicular directions, so that its two members retain the directional information that I_1 discards. The second shows that a cofactor-based directional measure reads the stretch in the perpendicular direction rather than introducing an independent third in-plane descriptor.

Objective directional descriptors solve only the first part of the design problem. Because the learned contribution $H(\mathbf{z})$ is nondecreasing in each feature, the feature vector $\mathbf{z}(\mathbf{F})$ also determines which deformation states it must order. For any two deformation states,

$$\mathbf{z}(\mathbf{F}_B) \geq \mathbf{z}(\mathbf{F}_A) \text{ componentwise} \implies H(\mathbf{z}(\mathbf{F}_B)) \geq H(\mathbf{z}(\mathbf{F}_A)), \quad (23)$$

whatever the width of the network. As anticipated in Section 1.2, this ordering constrains the learned contribution but not the analytic normalization and growth terms, so the feature map must be designed

together with them. Additional features remain functions of the same three engineering-strain components and introduce no new strain variable, but they embed the strain state into more variables on which the core is monotone and convex. They can enlarge the admissible response family without guaranteeing that it contains the target response.

The paired construction below is designed to retain directional stretch, reshape the componentwise ordering seen by the monotone core, and remain convex in the independent variables $(\mathbf{F}, J)$. It meets these three objectives by coupling directional stretch to volume through admissible determinant exponents. A feature can consequently decrease along a path when its denominator grows sufficiently, even if a numerator increases. Its reference derivative is also controlled analytically, which will allow the reference stress to be removed without sacrificing convexity.

3.3.1 Paired directional features

Let m denote the number of scalar features. For each $i = 1, \dots, m$, let $\mathbf{d}_i = (\cos \theta_i, \sin \theta_i)^T$ and $\mathbf{d}_i^\perp = (-\sin \theta_i, \cos \theta_i)^T$. Define

$$\begin{aligned} z_i(\mathbf{F}, J) &= \frac{I_4(\mathbf{d}_i)^{p_i}}{p_i J^{b_i}} + \frac{I_4(\mathbf{d}_i^\perp)^{q_i}}{q_i J^{c_i}} \\ &= \frac{\|\mathbf{F}\mathbf{d}_i\|^{2p_i}}{p_i J^{b_i}} + \frac{\|\mathbf{F}\mathbf{d}_i^\perp\|^{2q_i}}{q_i J^{c_i}}, \end{aligned} \quad (24)$$

where θ_i sets the orientation of the reference directions, p_i, q_i are the powers of the squared-stretch invariants, and b_i, c_i control each contribution's coupling to the area ratio J . These roles provide expressiveness, but the exponents cannot be chosen arbitrarily if the feature is to remain convex. They satisfy

$$p_i, q_i \geq \frac{1}{2}, \quad 0 \leq b_i \leq 2p_i - 1, \quad 0 \leq c_i \leq 2q_i - 1. \quad (25)$$

The lower bounds on p_i and q_i make the corresponding powers of the directional norms convex, while the upper bounds on b_i and c_i prevent their coupling with J from introducing a negative-curvature direction. Appendix B derives these conditions from the Hessian of a single stretch-determinant contribution. In the learned-feature variants, θ_i and the four exponent parameters are optimized, with p_i, q_i, b_i, c_i subject to Eq. (25). The fixed-feature controls of Section 5.2 hold the same initialized feature parameters constant while training the core. With unconstrained parameters $\hat{p}_i, \hat{q}_i, \hat{b}_i, \hat{c}_i \in \mathbb{R}$ and the functions $\text{softplus}(x) = \log(1 + e^x)$ and $\text{sigmoid}(x) = (1 + e^{-x})^{-1}$, whose ranges are $(0, \infty)$ and $(0, 1)$, the exponents are

$$\begin{aligned} p_i &= \frac{1}{2} + \text{softplus}(\hat{p}_i), & b_i &= (2p_i - 1) \text{sigmoid}(\hat{b}_i), \\ q_i &= \frac{1}{2} + \text{softplus}(\hat{q}_i), & c_i &= (2q_i - 1) \text{sigmoid}(\hat{c}_i). \end{aligned} \quad (26)$$

The angles require no admissibility bound, because the maps $\mathbf{F} \mapsto \mathbf{F}\mathbf{d}_i$ and $\mathbf{F} \mapsto \mathbf{F}\mathbf{d}_i^\perp$ remain linear in $\mathbf{F}$ for any fixed θ_i . Equation (26) satisfies the bounds of Eq. (25) strictly for all finite parameter values, approaching the boundary only as a limit, so no penalty is needed during training. Each z_i is one scalar feature with two perpendicular contributions. Because their exponents need not be equal, the pairing does not impose invariance under their interchange.

The construction was also designed to simplify reference normalization. Its paired and scaled contributions make the reference strain gradient of every feature isotropic. Consequently, the learned contribution has an isotropic reference stress, which a single scalar determinant correction can cancel in Section 3.5. The following result gathers the objectivity, convexity, and reference-state properties needed there.

Proposition 1. *Under the constraints in Eq. (25), each feature in Eq. (24) is objective and convex in the independent variables $(\mathbf{F}, J)$ on $J > 0$. Its value and engineering-strain gradient at the identity, evaluated with $J = \det \mathbf{F}$, are*

$$z_i^0 = \frac{1}{p_i} + \frac{1}{q_i}, \quad \partial_{\mathbf{e}} z_i|_{\mathbf{0}} = \rho_i(1, 1, 0)^T, \quad \rho_i = 2 - \frac{b_i}{p_i} - \frac{c_i}{q_i}. \quad (27)$$

The proof is given in Appendix B. The mechanism behind the reference gradient is that $1/p_i$ and $1/q_i$ cancel the stretch-power multipliers upon differentiation. The two perpendicular directional contributions then sum to $2\mathbf{I}$, while the determinant terms contribute $-(b_i/p_i + c_i/q_i)\mathbf{I}$. Hence ρ_i is the amplitude of an isotropic reference gradient, with equal normal components and zero shear. This reference property does not imply isotropy of the response away from the reference state.

The raw features are now mechanically admissible, but their numerical magnitudes may differ. Before the features enter the core, they are centered and scaled as $y_i = (z_i - z_i^{\text{cen}})/\tau_i$, with a centering value z_i^{cen} and a scale $\tau_i > 0$ that are independent of deformation for any fixed parameter set. This positive affine conditioning preserves both convexity in independent $(\mathbf{F}, J)$ and the componentwise ordering required by the monotone core. The choices used in the constitutive assessments are specified in Section 5.1.1.

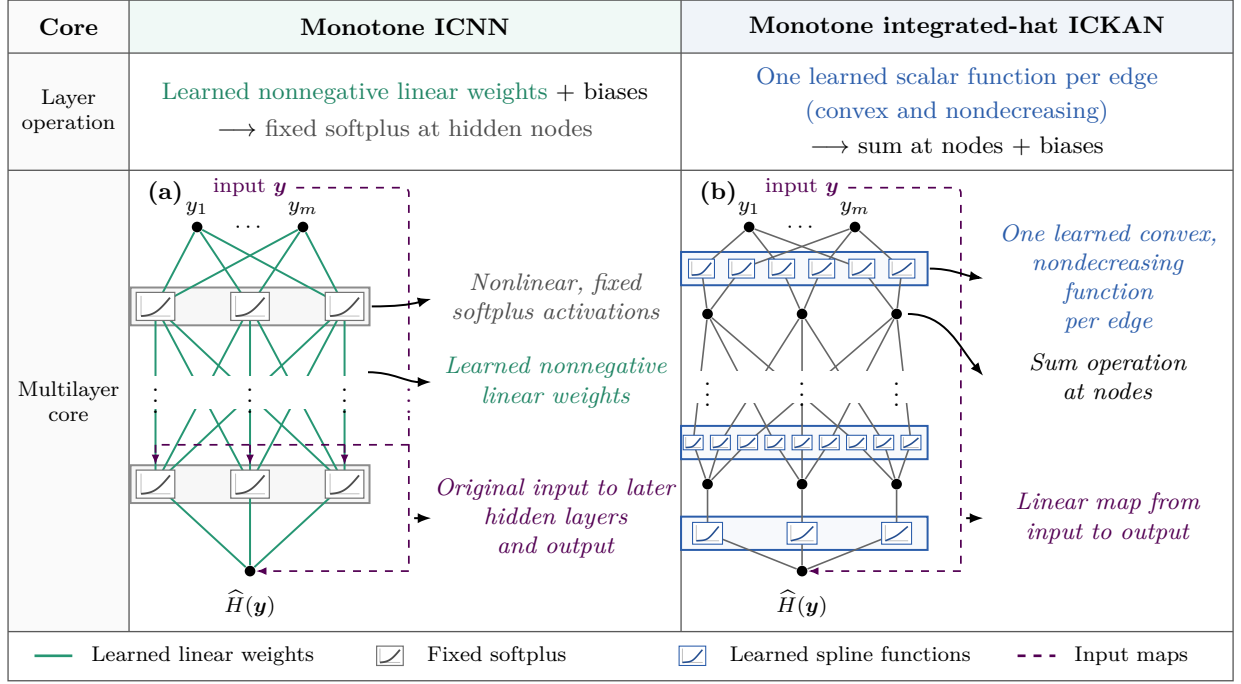


Figure 2: Multilayer monotone cores mapping the conditioned paired features $\mathbf{y}$ to the scalar contribution $\hat{H}(\mathbf{y})$. (a) The ICNN learns nonnegative linear maps and applies fixed softplus activations at hidden nodes. (b) The ICKAN learns one convex, nondecreasing scalar function per edge and sums them at nodes. Purple dashed paths denote nonnegative maps of the complete input, and ellipses denote intervening layers. Glyph counts indicate where each operation acts, not deployed widths or parameter counts. Unrestricted biases are omitted.

3.4 Monotone ICNN and integrated-hat ICKAN cores

Section 3.3 supplies features z_i that are objective and convex in the independent variables $(\mathbf{F}, J)$. The remaining question is how a learned core may combine them, through the conditioned vector $\mathbf{y} = (y_1, \dots, y_m)^T$, without undoing that convexity.

Composition with the convex feature map preserves convexity only when the core is convex and componentwise nondecreasing [13, 14]. The two cores enforce these properties differently. The ICNN learns nonnegative linear maps and applies fixed softplus activations at its hidden nodes, whereas the ICKAN learns a convex, nondecreasing scalar function on every edge and sums the incoming contributions at each node (Fig. 2).

3.4.1 Monotone ICNN

Let L denote the number of hidden layers and $\mathbf{a}^{(\ell)}$ the vector of activations in layer ℓ , for $\ell = 1, \dots, L$. Here $\hat{H}(\mathbf{y})$ denotes the scalar neural core acting on the conditioned features $\mathbf{y}$, to which the complete energy of Section 3.5 adds the reference-normalization and growth terms. The deployed ICNN has the form

$$\begin{aligned}
 \mathbf{a}^{(1)} &= \phi(\mathbf{U}_1 \mathbf{y} + \mathbf{b}_1), \\
 \mathbf{a}^{(\ell)} &= \phi(\mathbf{W}_\ell \mathbf{a}^{(\ell-1)} + \mathbf{U}_\ell \mathbf{y} + \mathbf{b}_\ell), \quad \ell = 2, \dots, L, \\
 \hat{H}(\mathbf{y}) &= \mathbf{v}^T \mathbf{a}^{(L)} + \mathbf{w}^T \mathbf{y} + b_{\text{out}},
 \end{aligned} \tag{28}$$

where $\phi = \text{softplus}$ is applied componentwise. Its derivatives satisfy $\phi'(x) = \text{sigmoid}(x) > 0$ and $\phi''(x) = \text{sigmoid}(x)[1 - \text{sigmoid}(x)] > 0$. The activation is smooth, increasing, and convex, and its smoothness makes the core C^2 , as required for a continuous constitutive tangent. The matrices $\mathbf{U}_\ell$ and $\mathbf{W}_\ell$ contain the weights from $\mathbf{y}$ and the preceding hidden layer, respectively, and $\mathbf{b}_\ell$ is the layer bias vector. The output weight vectors $\mathbf{v}$ and $\mathbf{w}$ act on $\mathbf{a}^{(L)}$ and $\mathbf{y}$, respectively, and b_{out} is an additive scalar output bias.

Every entry of $\mathbf{W}_\ell, \mathbf{U}_\ell, \mathbf{v}, \mathbf{w}$ is nonnegative, enforced by softplus transformations of unconstrained weight parameters. These signs play two related but distinct roles. The matrices $\mathbf{W}_\ell$ and the output vector $\mathbf{v}$ multiply convex hidden activations, so their nonnegativity preserves convexity as well as monotonicity. The maps $\mathbf{U}_\ell \mathbf{y}$ and $\mathbf{w}^T \mathbf{y}$ are affine, so they leave convexity unchanged for either sign, but nonnegative $\mathbf{U}_\ell$ and $\mathbf{w}$ are required for componentwise monotonicity. Biases are unrestricted because adding constants to a

preactivation preserves both properties. Hidden biases can change the core’s derivatives with respect to $\mathbf{y}$, whereas b_{out} only shifts $\hat{H}$ and leaves its derivatives unchanged.

The preservation argument proceeds layer by layer. The first hidden layer is convex because softplus is convex and its argument is affine, and it is nondecreasing because $\mathbf{U}_1$ is nonnegative. Assuming that $\mathbf{a}^{(\ell-1)}$ is convex and nondecreasing, the nonnegative map $\mathbf{W}_\ell \mathbf{a}^{(\ell-1)}$, the nonnegative affine input map, and composition with ϕ preserve both properties in layer ℓ . The same argument applies to the output, proving by induction that $\hat{H}$ is convex and componentwise nondecreasing. This is the ICNN composition argument of Amos et al. [42], supplemented here by nonnegative direct-input maps to guarantee monotonicity.

3.4.2 Monotone integrated-hat ICKAN

With $\mathbf{h}^{(0)} = \mathbf{y}$ and n_ℓ the width of layer ℓ , where $n_0 = m$, the hidden layers and scalar output of the ICKAN are

$$\begin{aligned} h_j^{(\ell)} &= \sum_{i=1}^{n_{\ell-1}} f_{ji}^{(\ell)}(h_i^{(\ell-1)}) + b_j^{(\ell)}, & j = 1, \dots, n_\ell, \quad \ell = 1, \dots, L, \\ \hat{H}(\mathbf{y}) &= \sum_{i=1}^{n_L} f_i^{\text{out}}(h_i^{(L)}) + \mathbf{w}^T \mathbf{y} + b_{\text{out}}, & \mathbf{w} \geq \mathbf{0}. \end{aligned} \quad (29)$$

Here $f_{ji}^{(\ell)}$ transforms the scalar carried by one edge and $h_j^{(\ell)}$ is the sum formed at a node. The nonnegative linear skip $\mathbf{w}^T \mathbf{y}$ connects the conditioned input directly to the scalar output. It remains to specify the functions appearing on the edges. For any $f_{ji}^{(\ell)}$ or f_i^{out} in Eq. (29), let x denote its scalar input. Each edge uses the common form

$$f(x) = ax + \sum_k c_k B_k(x), \quad a, c_k \geq 0, \quad (30)$$

where a is the learned linear slope, c_k are learned weights of the spline bases B_k , and k indexes these bases. Softplus transformations of unconstrained parameters impose the signs of a and c_k . Additive constants are represented once, through the unrestricted node biases in Eq. (29), rather than redundantly on every incoming edge.

To exclude negative slope and curvature by construction, the curvature of each edge is parameterized directly. Each B_k' is a nonnegative triangular hat centered at a knot κ_k . The knots form a fixed uniform grid with spacing $h > 0$ that is not optimized. Writing $s = (x - \kappa_k)/h$, one curvature basis is

$$B_k''(x) = \max(1 - |s|, 0). \quad (31)$$

Integrating twice from the left conditions $B_k(x) = B_k'(x) = 0$ for $x \leq \kappa_k - h$ gives

$$B_k(x) = h^2 \begin{cases} 0, & s \leq -1, \\ (s+1)^3/6, & -1 < s \leq 0, \\ (-s^3 + 3s^2 + 3s + 1)/6, & 0 < s < 1, \\ s, & s \geq 1. \end{cases} \quad (32)$$

Figure 3 shows both levels of the construction: integration of one curvature basis and assembly of a learned edge from several bases.

The value and first two derivatives in Eq. (32) match at every join, and B_k'' vanishes at the outer joins, so B_k is globally C^2 . Its curvature is localized to $[\kappa_k - h, \kappa_k + h]$, whereas the integrated basis has a zero left tail and a linear right tail. Moreover, $B_k'(-\infty) = 0$ and $B_k'' \geq 0$ imply $B_k' \geq 0$. The complete edge therefore satisfies

$$f''(x) = \sum_k c_k B_k''(x) \geq 0, \quad f'(x) = a + \sum_k c_k B_k'(x) \geq 0, \quad (33)$$

for every learned parameter set with $a, c_k \geq 0$, which makes each edge convex and nondecreasing by construction.

Because every edge is convex and nondecreasing, induction across the layers of Eq. (29) shows that every $h_j^{(\ell)}$, and finally $\hat{H}$, is convex and componentwise nondecreasing in $\mathbf{y}$. The nonnegative input-to-output skip preserves both properties. The fixed knot grid and finite number of bases limit where an individual edge can learn curvature, since outside their combined support the edge has linear tails. These tails are linear in the scalar input x only, and the complete energy remains nonlinear in $\mathbf{F}$ through the features and the analytic growth terms. The grid used in the constitutive assessment is reported in Section 5.1.1. This core is inspired by the monotonic input-convex KAN framework [48], with the integrated-hat bases and analytic tails specified here.

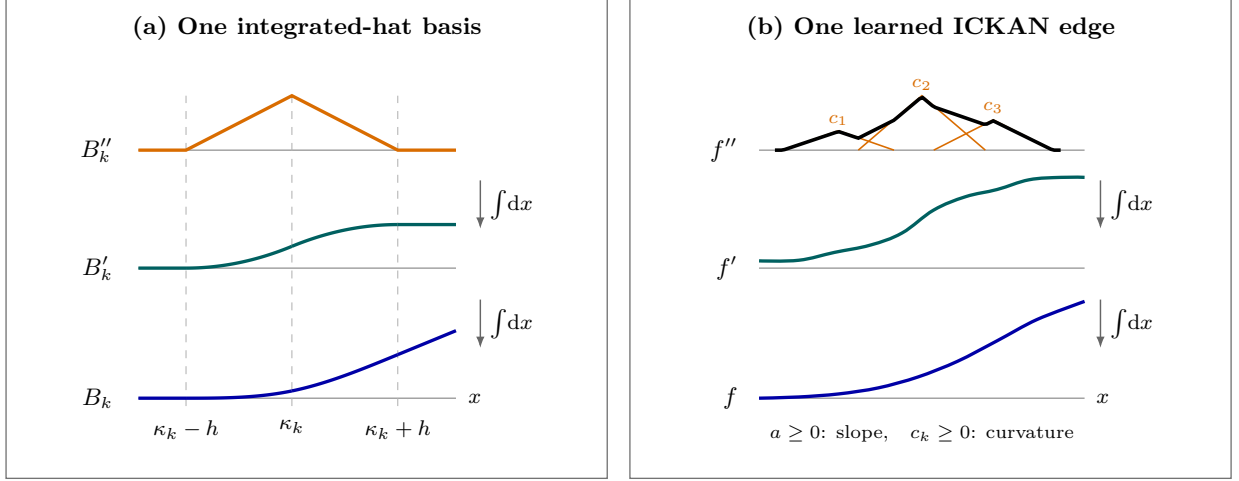


Figure 3: Integrated-hat parameterization of an ICKAN edge. (a) A localized nonnegative curvature hat integrates to a nonnegative slope and a convex C^2 basis with a zero left tail and a linear right tail. (b) Nonnegative learned coefficients combine several fixed bases into a nonnegative edge curvature, and integration gives a nondecreasing slope and a convex, nondecreasing connection. The curves illustrate the analytic construction, not the deployed grid.

3.5 The complete energy and its admissibility guarantees

Sections 3.3 and 3.4 constructed the physical feature vector $\mathbf{z}$, its conditioned representation $\mathbf{y}(\mathbf{z})$, and the scalar neural core $\hat{H}(\mathbf{y})$, which together define the learned contribution

$$H(\mathbf{z}) = \hat{H}(\mathbf{y}(\mathbf{z})). \quad (34)$$

This term represents the anisotropic response. The energy and stress references and the growth toward the boundary of the admissible deformation domain are set by analytic terms.

The residual stress produced by H at the undeformed state is quantified first. With $\mathbf{z}^0 = \mathbf{z}(\mathbf{I}, 1)$ and the feature-gradient factors ρ_i of Eq. (27), the core sensitivities and the scalar amplitude of this residual stress are

$$h_i = \partial_{z_i} H(\mathbf{z}^0) \geq 0, \quad r = \sum_i h_i \rho_i. \quad (35)$$

For a fixed parameter set, $H(\mathbf{z}^0)$ and r are constants with respect to deformation. The complete stored energy then combines the learned contribution with analytic reference-normalization and growth terms,

$$\begin{aligned} W(\mathbf{F}) = & \underbrace{H(\mathbf{z}) - H(\mathbf{z}^0)}_{\text{(i) } W_{\text{learn}}} + \underbrace{[-r(J-1)]}_{\text{(ii) } W_{\text{corr}}} \\ & + \underbrace{\alpha(J-1 - \log J) + \frac{\beta}{2}(J-1)^2}_{\text{(iii) } W_{\text{vol}}} \\ & + \underbrace{\varepsilon \left(\frac{\text{tr } \mathbf{C}}{2} - 1 - \log J \right)}_{\text{(iv) } W_{\text{dist}}}, \quad \alpha, \beta, \varepsilon > 0. \end{aligned} \quad (36)$$

Here $J = \det \mathbf{F}$, $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, and $\mathbf{z} = \mathbf{z}(\mathbf{F}, J)$. The positive energy-density coefficients α , β , and ε weight the analytic growth terms. For either core, the composition in Eq. (34) is twice continuously differentiable, convex, and componentwise nondecreasing in $\mathbf{z}$.

Each of the four contributions named by the underbraces solves a distinct constitutive problem:

- (i) The term W_{learn} carries the learned response and fixes the energy reference. At $\mathbf{F} = \mathbf{I}$, the features equal $\mathbf{z}^0$, and hence

$$W(\mathbf{I}) = H(\mathbf{z}^0) - H(\mathbf{z}^0) = 0. \quad (37)$$

The remaining terms also vanish because $J = 1$ and $\text{tr } \mathbf{C} = 2$. Since $H(\mathbf{z}^0)$ is constant with respect to deformation, its subtraction fixes the energy reference without changing the stress or tangent.

- (ii) The correction W_{corr} removes the reference stress, which a constant subtraction cannot change. By the chain rule and Eq. (27), the learned contribution has the reference value

$$\nabla_{\mathbf{e}} H(\mathbf{z})|_0 = \sum_i h_i \nabla_{\mathbf{e}} z_i|_0 = \sum_i h_i \rho_i (1, 1, 0)^T = r(1, 1, 0)^T. \quad (38)$$

The identity $J = \sqrt{(1 + 2E_{11})(1 + 2E_{22}) - (2E_{12})^2}$ gives $\nabla_{\mathbf{e}} J|_0 = (1, 1, 0)^T$. Holding r fixed when differentiating yields

$$\nabla_{\mathbf{e}} [H(\mathbf{z}) - r(J - 1)]|_0 = r(1, 1, 0)^T - r(1, 1, 0)^T = \mathbf{0}. \quad (39)$$

A single scalar r suffices because the paired features make the reference stress isotropic (Proposition 1). The correction is zero in value at $J = 1$ and, being affine in the independent determinant variable, preserves convexity for either sign of r . Linden et al. [14] use the same affine determinant term to normalize isotropic energies but need additional invariant terms for anisotropic ones, whereas Eq. (27) makes the isotropic form sufficient for the paired features.

- (iii) The volumetric term W_{vol} controls growth in J . Its logarithmic part diverges as $J \rightarrow 0^+$, where the quadratic part alone would remain finite with limit $\beta/2$. The quadratic part dominates the linear correction $-r(J - 1)$ as $J \rightarrow \infty$, so the energy diverges under expansion for either sign of r . At the reference,

$$W_{\text{vol}}(1) = 0, \quad W'_{\text{vol}}(J) = \alpha(1 - J^{-1}) + \beta(J - 1), \quad W'_{\text{vol}}(1) = 0, \quad (40)$$

so this term leaves the reference energy and stress unchanged.

- (iv) The distortional term W_{dist} supplies growth under area-preserving changes of shape, which a function of J alone cannot detect. For example, with $\lambda > 0$,

$$\mathbf{F} = \text{diag}(\lambda, \lambda^{-1}), \quad J = 1, \quad \varepsilon \left(\frac{\text{tr} \mathbf{C}}{2} - 1 - \log J \right) = \frac{\varepsilon}{2} (\lambda^2 + \lambda^{-2} - 2). \quad (41)$$

This contribution diverges as $\lambda \rightarrow \infty$, although the area remains constant. More generally, $\text{tr} \mathbf{C} = \|\mathbf{F}\|_F^2$ supplies growth under unbounded deformation at bounded positive J . At the reference, the constant -1 sets its value to zero. Using $\mathbf{C} = \mathbf{I} + 2\mathbf{E}$ and $\partial_{\mathbf{E}} J|_0 = \mathbf{I}$, its stress contribution is

$$\partial_{\mathbf{E}} \left[\varepsilon \left(\frac{\text{tr} \mathbf{C}}{2} - 1 - \log J \right) \right] \Big|_0 = \varepsilon [\mathbf{I} - J^{-1} \partial_{\mathbf{E}} J|_0] = \mathbf{0}. \quad (42)$$

Together, W_{learn} and W_{corr} set the learned response and its reference state, while W_{vol} and W_{dist} prescribe growth toward the boundary of the admissible domain. Although the analytic terms vanish with their first derivatives at the reference, they still contribute to the reference stiffness, because J is nonlinear in $\mathbf{e}$. The following proposition gathers the guarantees of their complete sum.

Proposition 2. *Let the features satisfy Eq. (25), let the core be either of the C^2 convex, componentwise nondecreasing constructions in Section 3.4, and let $\alpha, \beta, \varepsilon > 0$. Then Eq. (36) defines a C^2 , objective, two-dimensional polyconvex energy on $\text{GL}^+(2)$. Its associated second Piola–Kirchhoff stress $\mathbf{S}$ is symmetric, its work-conjugate representation is $\mathbf{s} = \nabla_{\mathbf{e}} W$, $W(\mathbf{I}) = 0$, $\mathbf{s}(\mathbf{0}) = \mathbf{0}$, and W diverges under collapse, unbounded expansion, and unbounded distortion at bounded positive area. Thus the construction satisfies (C1)–(C6).*

The proof is given in Appendix B.3. In brief, the convex, componentwise nondecreasing core preserves feature convexity, the remaining terms are constant, affine, or convex in independent $(\mathbf{F}, J)$, and the reference identities and growth limits complete the argument.

The three constitutive constructions carried into the numerical assessment are summarized in Table 1. The ICNN and ICKAN columns include both fixed- and learned-feature variants, and learning the feature parameters changes the representation while leaving the guarantees above intact.

Table 1: Constitutive constructions carried into the numerical assessment. Criteria (C1)–(C6) are defined in Section 3.1, and the guarantees refer to the constructions of this section.

	Unconstrained energy	ICNN	ICKAN
Role	Flexible hyperelastic reference	Guaranteed paired-feature energy	Guaranteed paired-feature energy
Inputs	Objective measures from $(\mathbf{C}, J)$	Paired features (fixed or learned)	Paired features (fixed or learned)
Learned scalar map	Smooth energy network	Nonnegative weights, fixed activations	Convex, nondecreasing edge functions
(C1)–(C4)	By construction	By construction	By construction
(C5)–(C6)	Not imposed	By construction	By construction

4 Projection-based reduced-order models

The learned energies of Section 3 remove the microscopic solve from online constitutive evaluation. To place their structural accuracy and cost against physics-based alternatives, Section 5.3.5 also uses three projection-based microscopic surrogates as intrusive references. These models instead approximate the microscopic displacement field while retaining the underlying constituent law. Reduced coordinates decrease the number of equilibrium unknowns, and hyperreduction decreases the number of element evaluations. The three references are an affine HPROM, an iterative HPROM–ANN with adaptive integration, and its direct counterpart D-HPROM–ANN.

Figure 4 summarizes the construction, from the offline RVE snapshots to the affine and nonlinear state representations and the three online routes, which share the same stress–tangent output.

4.1 Periodic microscopic reference

All three reduced evaluators derive from the periodic RVE problem of Section 2, whose N -dimensional discretization defines the FOM. A PROM solves the same equilibrium problem in reduced coordinates, and an HPROM additionally hyperreduces its projected operators.

The reduced models use the independent state $\mathbf{u} \in \mathbb{R}^N$ obtained from the physical nodal displacement vector $\mathbf{d} \in \mathbb{R}^{N_{\text{dof}}}$ after enforcing periodicity and removing rigid translation. In the deployed convention,

$$\mathbf{d}(\mathbf{u}, \mathbf{e}) = \mathbf{T}\mathbf{u} + \mathbf{g}_{\text{per}}(\mathbf{e}), \quad \mathbf{R}(\mathbf{u}; \mathbf{e}) = \mathbf{T}^T \mathbf{f}_{\text{int}}(\mathbf{d}(\mathbf{u}, \mathbf{e})) = \mathbf{0}. \quad (43)$$

Here $\mathbf{T}$ identifies periodic degrees of freedom, and the lifting $\mathbf{g}_{\text{per}}$ supplies the strain-dependent displacement jumps and anchor values of Section 2, so any independent state $\mathbf{u}$ produces a physical field that satisfies the prescribed kinematic constraints. With $\mathbf{K}_{\text{phys}} = \partial_{\mathbf{d}} \mathbf{f}_{\text{int}}$ the assembled physical stiffness, the FOM tangent is $\mathbf{K} = \partial_{\mathbf{u}} \mathbf{R} = \mathbf{T}^T \mathbf{K}_{\text{phys}} \mathbf{T}$, and all residuals follow the internal-force sign convention.

The reduced bases are built from snapshots of this affine-retaining state, which relates to the fluctuation coordinates of Eq. (8) through $\mathbf{a} = \mathbf{u} - \mathbf{u}_{\text{aff}}(\mathbf{e})$, with $\mathbf{u}_{\text{aff}}$ the affine displacement at the independent nodes. This choice leaves the periodic boundary conditions unchanged and keeps the load-correlated information used by the strain-informed identification below, so the reported POD dimension characterizes this representation and its truncation tolerance.

4.2 POD approximation and the affine PROM

The first reduction targets the number of equilibrium unknowns. FOM solutions $\mathbf{u}^s$ at sampled strains $\mathbf{e}^s$, $s = 1, \dots, N_s$, are centered about a reference state $\mathbf{u}_{\text{ref}}$, and the singular value decomposition (SVD) of the snapshot matrix

$$\mathbf{S}_{\text{snap}} = [\mathbf{u}^1 - \mathbf{u}_{\text{ref}}, \dots, \mathbf{u}^{N_s} - \mathbf{u}_{\text{ref}}] = \mathbf{U}_S \mathbf{\Sigma}_S \mathbf{Y}_S^T. \quad (44)$$

supplies the POD modes. The n_{tra} leading left singular vectors form the orthonormal reduced-order basis (ROB) $\mathbf{V}_{\text{tra}}$ of the affine approximation

$$\tilde{\mathbf{u}} = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tra}} \mathbf{q}_{\text{tra}}, \quad \mathbf{q}_{\text{tra}} \in \mathbb{R}^{n_{\text{tra}}}, \quad \mathbf{V}_{\text{tra}}^T \mathbf{V}_{\text{tra}} = \mathbf{I}. \quad (45)$$

which replaces the N independent displacement coordinates by $n_{\text{tra}} \ll N$ modal amplitudes. The subscript “tra” marks this traditional affine approximation, and the term affine refers only to the approximation space, since the constitutive law and the reduced equilibrium equations remain nonlinear.

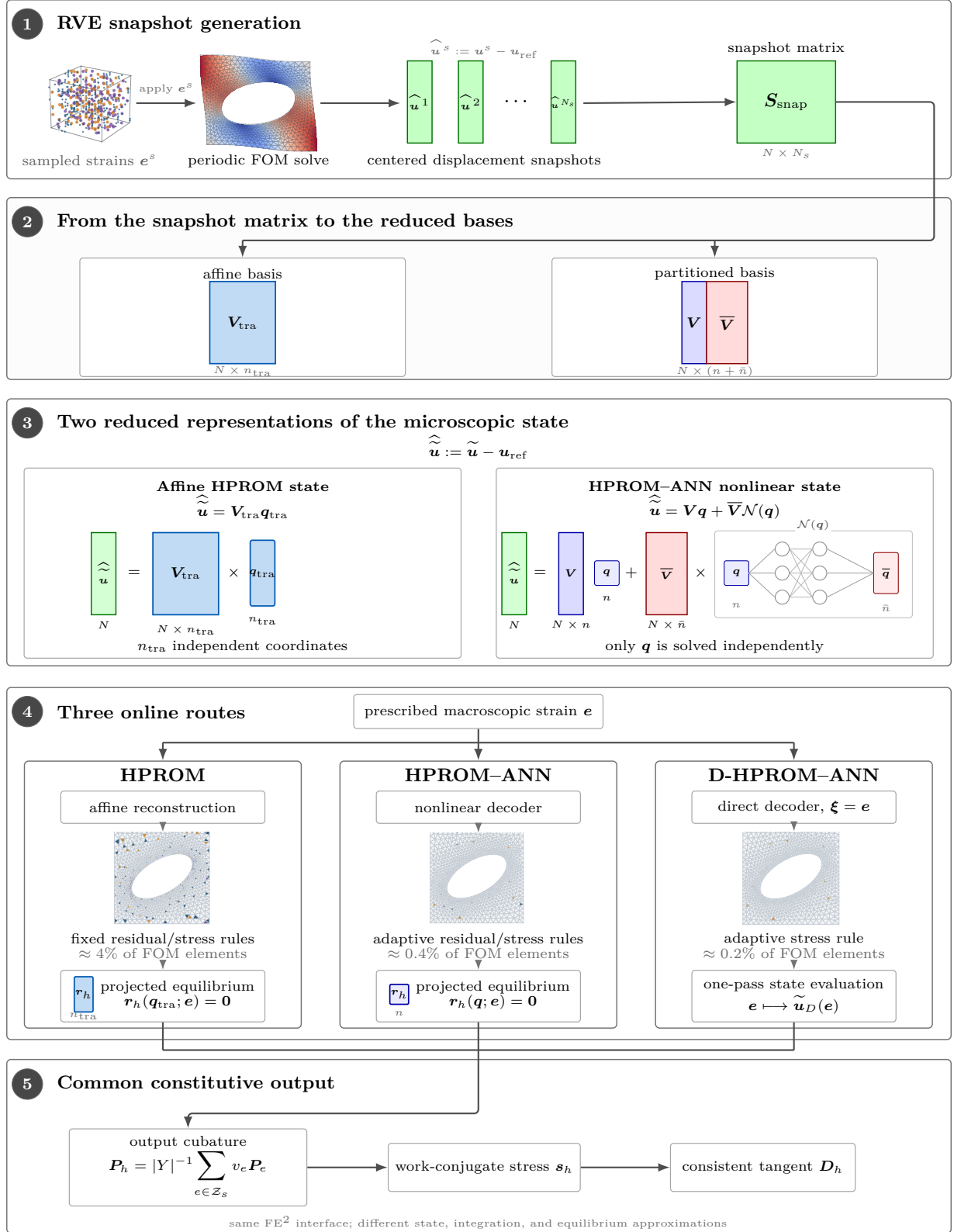


Figure 4: Intrusive reduced references used in the structural comparison. FOM snapshots provide either an affine state representation or a primary–secondary partition in which the ANN predicts the secondary coordinates. Online, HPROM and HPROM–ANN solve projected equilibrium with fixed and adaptive cubature, respectively, whereas D-HPROM–ANN evaluates the nonlinear decoder directly. All three retain microscopic constitutive evaluations and return the same stress–tangent interface.

The retained dimension is selected from the relative squared reconstruction error, which for a Euclidean POD is

$$\epsilon_{\text{POD}}^2(n_{\text{tra}}) = \frac{\sum_{j>n_{\text{tra}}} \sigma_j^2}{\sum_j \sigma_j^2}. \quad (46)$$

This criterion measures how well the snapshots are compressed but does not bound the stress error at unsampled strains, because force evaluation, constitutive differentiation, and equilibrium conditioning can make small displacement components important.

Galerkin projection onto the admissible variations $\delta \tilde{\mathbf{u}} = \mathbf{V}_{\text{tra}} \delta \mathbf{q}_{\text{tra}}$ gives the reduced equations and tangent at a prescribed strain $\mathbf{e}$,

$$\mathbf{r}_{\text{tra}}(\mathbf{q}_{\text{tra}}; \mathbf{e}) = \mathbf{V}_{\text{tra}}^T \mathbf{R}(\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tra}} \mathbf{q}_{\text{tra}}; \mathbf{e}) = \mathbf{0}, \quad \mathbf{K}_{\text{tra}} = \mathbf{V}_{\text{tra}}^T \mathbf{K} \mathbf{V}_{\text{tra}}. \quad (47)$$

The system now has n_{tra} unknowns, but evaluating $\mathbf{R}$ and $\mathbf{K}$ still visits every microscopic element, which the hyperreduction of Section 4.4 addresses.

4.3 Nonlinear reduction through latent-space closure

The POD dimension required to reconstruct a displacement field can exceed the number of coordinates needed to parametrize the loading family. Latent-space closure exploits this distinction by retaining a rich linear span while learning dependencies among its modal amplitudes, so that only a smaller subset remains unknown in equilibrium [22, 24]. An $n_{\text{tot}} = n + \bar{n}$ dimensional ROB is partitioned into orthonormal primary and secondary blocks,

$$\mathbf{V}_{\text{tot}} = [\mathbf{V}, \bar{\mathbf{V}}], \quad \mathbf{V}^T \mathbf{V} = \mathbf{I}, \quad \bar{\mathbf{V}}^T \bar{\mathbf{V}} = \mathbf{I}, \quad \mathbf{V}^T \bar{\mathbf{V}} = \mathbf{0}, \quad (48)$$

and each snapshot is projected onto the corresponding coordinates,

$$\mathbf{q}^s = \mathbf{V}^T (\mathbf{u}^s - \mathbf{u}_{\text{ref}}), \quad \bar{\mathbf{q}}^s = \bar{\mathbf{V}}^T (\mathbf{u}^s - \mathbf{u}_{\text{ref}}). \quad (49)$$

The primary coordinates $\mathbf{q} \in \mathbb{R}^n$ remain independent unknowns, whereas an ANN closure $\mathcal{N} : \mathbb{R}^n \rightarrow \mathbb{R}^{\bar{n}}$ predicts the secondary coordinates. The resulting nonlinear decoder and its tangent are

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \mathcal{N}(\mathbf{q}; \boldsymbol{\eta}), \quad \mathbf{B}(\mathbf{q}) = \partial_{\mathbf{q}} \tilde{\mathbf{u}} = \mathbf{V} + \bar{\mathbf{V}} \mathcal{N}_{,\mathbf{q}}. \quad (50)$$

The ANN predicts modal amplitudes of the microscopic displacement, not stress or energy, and the secondary modes remain in that field without appearing as independent equilibrium unknowns.

The closure is trained on snapshot pairs $(\mathbf{q}^s, \bar{\mathbf{q}}^s)$ by minimizing

$$\mathcal{L}_{\text{cl}}(\boldsymbol{\eta}) = \frac{1}{|\mathcal{I}_{\text{fit}}| \bar{n}} \sum_{s \in \mathcal{I}_{\text{fit}}} \|\mathcal{N}(\mathbf{q}^s; \boldsymbol{\eta}) - \bar{\mathbf{q}}^s\|_2^2. \quad (51)$$

Input standardization is absorbed into the closure map, while the secondary outputs remain unscaled. Smooth tanh activations provide the decoder derivatives required by projection. This coordinate loss measures displacement reconstruction within the secondary space.

The leading POD coefficients need not provide a well-conditioned primary-to-secondary map. The input-informed construction of Hernández et al. [39] instead combines the retained modal amplitudes linearly into three primary coordinates $\boldsymbol{\xi}$ approximately aligned with the three macroscopic strain components. If $\mathbf{A}_m$ denotes the resulting nonsingular coordinate transformation, the deployed decoder is

$$\tilde{\mathbf{u}}(\boldsymbol{\xi}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{A}_m \boldsymbol{\xi} + \bar{\mathbf{V}} \hat{\mathcal{N}}(\boldsymbol{\xi}), \quad \mathbf{B}_{\boldsymbol{\xi}} = \partial_{\boldsymbol{\xi}} \tilde{\mathbf{u}}. \quad (52)$$

Because the alignment $\boldsymbol{\xi}^s \approx \mathbf{e}^s$ is only fitted from the snapshots, the iterative PROM-ANN solves at fixed $\mathbf{e}$ for the state satisfying

$$\mathbf{r}(\boldsymbol{\xi}; \mathbf{e}) = \mathbf{B}_{\boldsymbol{\xi}}^T \mathbf{R}(\tilde{\mathbf{u}}(\boldsymbol{\xi}); \mathbf{e}) = \mathbf{0}, \quad (53)$$

rather than imposing $\boldsymbol{\xi} = \mathbf{e}$. Because $\mathbf{B}_{\boldsymbol{\xi}}$ depends on the state, the exact Newton derivative includes the curvature of the decoder manifold. Appendix D gives the coordinate construction, the reconstruction-error split, and this tangent explicitly.

For the comparison below, the affine and nonlinear models share the same retained span, $n_{\text{tot}} = n_{\text{tra}}$, and the nonlinear model reduces the number of equilibrium unknowns from n_{tra} to $n = 3$ before any hyperreduction. For compactness, the cubature and output formulas below retain $\mathbf{q}$ for the primary modal amplitudes, and the deployed nonlinear model evaluates the decoder and adaptive-weight maps through $\boldsymbol{\xi} = \mathbf{A}_m^{-1} \mathbf{q}$.

4.4 Hyperreduction of the projected equations

Hyperreduction replaces each full element sum in the projected equations with a sparse weighted sum, a second approximation introduced after the displacement representation has been chosen. Two integration tasks must be distinguished: assembling the projected equilibrium equations and recovering the homogenized stress. Hernández et al. [39] formulate both tasks and discuss the use of either a common or a dedicated output rule. Separate elementwise rules are used here because the two integrands and their accuracy requirements differ.

4.4.1 Fixed elementwise cubature

Let $\mathcal{E}$ be the microscopic element set and $\mathbf{L}_e$ extract element nodal values from $\mathbf{d}$. At fixed macroscopic input, define the reconstructed element state and its reduced tangent by

$$\mathbf{d}_e = \mathbf{L}_e(\mathbf{T}\tilde{\mathbf{u}} + \mathbf{g}_{\text{per}}), \quad \mathbf{B}_e = \mathbf{L}_e \mathbf{T} \mathbf{B}. \quad (54)$$

The full reduced residual is $\sum_{e \in \mathcal{E}} \mathbf{h}_e$, where $\mathbf{h}_e = \mathbf{B}_e^T \mathbf{f}_{\text{int},e}$. Fixed ECM approximates it as

$$\mathbf{r}_{\text{ECM}}(\mathbf{q}; \mathbf{e}) = \sum_{e \in \mathcal{Z}_r} w_e \mathbf{h}_e(\mathbf{q}; \mathbf{e}) = \mathbf{0}, \quad \mathcal{Z}_r \subset \mathcal{E}, \quad w_e \geq 0. \quad (55)$$

For the affine HPROM, which contains no ANN, $\mathbf{B}$ is replaced by $\mathbf{V}_{\text{tra}}$ and the unknown is $\mathbf{q}_{\text{tra}}$. A selected element retains its ordinary within-element quadrature, so selected elements, selected Gauss points, and independent coordinates are distinct counts.

Offline, the element contributions are collected over N_c training states and compressed by SVD. ECM then selects a sparse support and nonnegative multipliers that reproduce the retained integration modes [34]. If $\mathbf{G}_{\text{cub}}$ denotes those modes augmented by the constant-integration constraint (Appendix E) and $\mathbf{b}_{\text{cub}} = \mathbf{G}_{\text{cub}} \mathbf{1}$, selection stops when

$$\frac{\|\mathbf{G}_{\text{cub}} \mathbf{w} - \mathbf{b}_{\text{cub}}\|_2}{\|\mathbf{b}_{\text{cub}}\|_2} \leq \varepsilon_{\text{ECM}}. \quad (56)$$

The tolerance ε_{SVD} controls the preceding integrand compression, whereas ε_{ECM} controls the sparse fit. For the residual rule, the element data are $\mathbf{h}_e$. For the output rule, they are the four components of the element contribution to the average first Piola–Kirchhoff stress, all four of which are retained because this tensor is not generally symmetric. Both rules impose $\sum_e w_e = N_{\text{el}}$, with physical quadrature measures already included in the element contributions. Their supports and weights remain fixed online.

4.4.2 Manifold-adaptive weights

A fixed rule must represent every sampled state with the same multipliers. MAW–ECM instead retains one common support but lets its weights vary over the reduced manifold [39]. Applied to the residual, this gives

$$\mathbf{r}_{\text{MAW}}(\mathbf{q}; \mathbf{e}) = \sum_{e \in \mathcal{Z}_r} w_e(\mathbf{q}) \mathbf{B}_e(\mathbf{q})^T \mathbf{f}_{\text{int},e}(\mathbf{d}_e) = \mathbf{0}. \quad (57)$$

Starting from a fixed ECM rule, the MAW–ECM pruning algorithm removes elements while redistributing nonnegative statewise weights, through the local and graph-regularized stages described by Hernández et al. [39]. Pruning identifies the support but not the continuous online map, and neural fields are used here for that final regression stage. Because the equilibrium and stress rules are separate, two maps, $\mathcal{N}_r$ and $\mathcal{N}_s$, are trained on independently pruned supports $\mathcal{Z}_r$ and $\mathcal{Z}_s$,

$$\begin{aligned} \mathbf{w}(\boldsymbol{\xi}) &= N_{\text{el}} \text{softmax}(\mathcal{N}_r(\boldsymbol{\xi}; \boldsymbol{\theta}_r)), & \sum_{e \in \mathcal{Z}_r} w_e &= N_{\text{el}}, \\ \mathbf{v}(\boldsymbol{\xi}) &= N_{\text{el}} \text{softmax}(\mathcal{N}_s(\boldsymbol{\xi}; \boldsymbol{\theta}_s)), & \sum_{e \in \mathcal{Z}_s} v_e &= N_{\text{el}}. \end{aligned} \quad (58)$$

Input standardization is absorbed into each network. Unlike the closure ANN $\mathcal{N}$, which predicts secondary displacement coordinates, these networks predict integration weights, fitted against full-mesh element sums at each decoded training state with the loss of Appendix E. The scaled softmax guarantees positivity and normalization, whereas integration accuracy is learned. Offline, the closure is trained first, the two supports are then selected and pruned, and the two weight fields are fitted last.

Online, the current reduced iterate is passed to both the closure and $\mathcal{N}_r$, and $\mathcal{N}_s$ supplies the output weights, with the supports and all network parameters fixed. Differentiating the adaptive residual produces

three contributions: element stiffness, decoder curvature, and weight variation. Only the first enters the microscopic modified-Newton iteration, which changes the convergence rate but not the converged state, whereas the consistent constitutive tangent differentiates the complete implemented residual and output, as described next. For the affine HPROM, the decoder curvature and weight-variation terms vanish, so its projected stiffness is the exact residual Jacobian.

To translate sparse cubature into wall-clock savings, each optimized reduced mesh contains only the selected elements and the nodes they require. The corresponding basis operators and periodic lifting are restricted to that support before online evaluation. Online cost therefore depends on the support size, the retained secondary dimension, the equilibrium iterations, and the derivative evaluation, and not only on the number of primary coordinates.

4.5 Constitutive output

4.5.1 Equilibrium-based stress and consistent tangent

The reduced microscopic solver must return the same work-conjugate stress and tangent required by the FE² problem. The final stage of Figure 4 summarizes this common output path. The separate output rule introduced above reconstructs the average first Piola–Kirchhoff stress as

$$\mathbf{P}_h(\mathbf{q}, \mathbf{e}) = \frac{1}{|Y|} \sum_{e \in \mathcal{Z}_s} v_e(\mathbf{q}) \sum_{g \in e} \omega_g \mathbf{P}_{\mu, g}(\mathbf{d}_e), \quad \mathbf{S}_h = \mathbf{F}^{-1} \mathbf{P}_h. \quad (59)$$

Here v_e is fixed for HPROM and state dependent for HPROM–ANN. Because the independently fitted output rule need not produce an exactly symmetric converted tensor, the implementation returns its symmetric work-conjugate components,

$$\mathbf{s}_h(\mathbf{q}, \mathbf{e}) = ((S_h)_{11}, (S_h)_{22}, \frac{1}{2}[(S_h)_{12} + (S_h)_{21}])^T, \quad (60)$$

and differentiates this actual output. The residual and stress supports may overlap. Their union defines the element data required by the iterative constitutive evaluation, whereas only $\mathcal{Z}_r$ enters the equilibrium solve.

Reduced equilibrium defines an implicit state $\mathbf{q}^*(\mathbf{e})$ and hence the constitutive map $\mathbf{s}_h(\mathbf{q}^*(\mathbf{e}), \mathbf{e})$. If the residual Jacobian is nonsingular, differentiating equilibrium gives

$$\mathbf{D}_h = \mathbf{s}_{h, \mathbf{e}} - \mathbf{s}_{h, \mathbf{q}} \mathbf{r}_{h, \mathbf{q}}^{-1} \mathbf{r}_{h, \mathbf{e}}. \quad (61)$$

The partial derivatives hold the other argument fixed and are evaluated at the converged state. They include the periodic lifting, stress conversion, decoder curvature, and, for adaptive rules, weight variation. Equation (61), rather than the modified-Newton matrix used to reach microscopic equilibrium, supplies the tangent returned to the structural solver. Appendix E records how these derivatives are evaluated in the two implementations.

4.5.2 Direct decoder evaluation: D-HPROM–ANN

When input-informed coordinates closely approximate the loading parameters, the decoder can also be evaluated without solving reduced equilibrium, as discussed by Hernández et al. [39] for their hyperelastic example. Here this defines a separate operating mode, D-HPROM–ANN, obtained by prescribing $\boldsymbol{\xi} = \mathbf{e}$ in Eq. (52),

$$\tilde{\mathbf{u}}_D(\mathbf{e}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{A}_m \mathbf{e} + \bar{\mathbf{V}} \hat{\mathcal{N}}(\mathbf{e}). \quad (62)$$

The decoded state is inserted directly into the adaptive stress rule, using the same closure and stress-weight field as the iterative model. Only the stress support is needed, since no residual cubature or reduced equilibrium Jacobian is evaluated. The tangent is obtained by central differences of the complete direct stress map, updating the decoder, weights, lifting, and stress conversion at each perturbed input.

The accuracy of D-HPROM–ANN depends on both the closure and the approximation $\boldsymbol{\xi} \approx \mathbf{e}$. Conversely, solving projected equilibrium corrects the primary state only within the chosen manifold and cubature approximation, so it need not give a smaller stress error than direct evaluation at every input.

4.6 Mechanical structure retained by the reduced models

Unlike a direct constitutive surrogate, every reduced model above continues to evaluate the original microscopic hyperelastic law, and the iterative variants also enforce projected equilibrium. This preserves a direct

micromechanical link, but it does not automatically make the deployed macroscopic stress the derivative of an effective potential.

For fixed weights and consistent kinematics, Galerkin internal forces can be derived from a weighted reduced energy, as in the structure-preservation results established for ECSW under their stated hypotheses [33]. Two features of the present deployment prevent that conclusion from transferring without further assumptions. First, equilibrium and stress use independently fitted cubature rules, so they need not derive from the same energy. Second, when the weights depend on the reduced state, weighting energy derivatives is not equivalent to differentiating the weighted energy, because

$$\nabla_{\mathbf{q}} \sum_e w_e(\mathbf{q}) \Pi_e(\mathbf{q}, \mathbf{e}) = \sum_e w_e(\mathbf{q}) \nabla_{\mathbf{q}} \Pi_e + \sum_e \Pi_e \nabla_{\mathbf{q}} w_e. \quad (63)$$

The final term is absent from Eq. (57), and nonnegative weights do not eliminate it. Consequently, no exact effective potential or polyconvexity certificate is claimed for the deployed HPROM variants, whose consistent tangent is the derivative of the implemented stress map and need not be symmetric.

Table 2 summarizes the three deployed evaluators and their FOM reference.

Table 2: Intrusive evaluators used in the structural accuracy–cost comparison and their FOM reference. All reduced models retain microscopic constitutive evaluations.

Model	State representation	Equilibrium unknowns	Integration	Learned component
FOM	$\mathbf{u}$	N	Full mesh	None
HPROM	$\tilde{\mathbf{u}} = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tra}} \mathbf{q}_{\text{tra}}$	n_{tra}	Fixed ECM	None
HPROM-ANN	$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \mathcal{N}(\mathbf{q})$	n	Adaptive ECM	Secondary coordinates; residual and stress weights
D-HPROM-ANN	$\tilde{\mathbf{u}}_D(\mathbf{e}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{A}_m \mathbf{e} + \bar{\mathbf{V}} \hat{\mathcal{N}}(\mathbf{e})$	None	Adaptive stress rule	Same closure and stress weights as HPROM-ANN

5 Numerical examples

We use two periodic microstructures, each with its own independently trained models, so that the assessment does not rest on a single cavity arrangement. The multicavity RVE (MC-RVE) presents several geometric directions within one cell, which makes it a natural setting in which to isolate the effect of learning the paired features and of changing their number. The single-cavity RVE (SC-RVE) is instead the microscopic model subsequently propagated through the structural FE² calculation and shared with the intrusive references of Section 4; it therefore supplies the deployment qualification. Together, the two benchmarks test the same constitutive construction in distinct geometric settings, but not transfer of parameters between them.

The evidence follows those roles in causal order. After the common setting, we first ask on the MC-RVE how many paired features are needed and what learning them adds. We then ask on the SC-RVE whether the selected learned energies are accurate enough for the structural calculation of Section 5.3.5. Throughout, analytical guarantees are distinguished from predictive accuracy and finite implementation checks.

5.1 Common setting

Both benchmarks are square periodic cells with total void fraction 0.20. Their solid phase follows the same compressible Neo-Hookean law in plane strain. With $\mathbf{C}_\mu = \mathbf{F}_\mu^T \mathbf{F}_\mu$ and $J_\mu = \det \mathbf{F}_\mu > 0$, its local energy density is

$$\psi_\mu = \frac{\mu}{2} (\text{tr } \mathbf{C}_\mu - 2) - \mu \log J_\mu + \frac{\lambda}{2} (\log J_\mu)^2, \quad (64)$$

with Young’s modulus 1.628 GPa and Poisson ratio 0.4, where μ and λ are the Lamé constants. The cavity geometry, rather than the matrix law, distinguishes the two benchmarks.

FOM microscopic fields at a common diagnostic state $(E_{11}, E_{22}, 2E_{12}) = (18, -4, 8)\%$

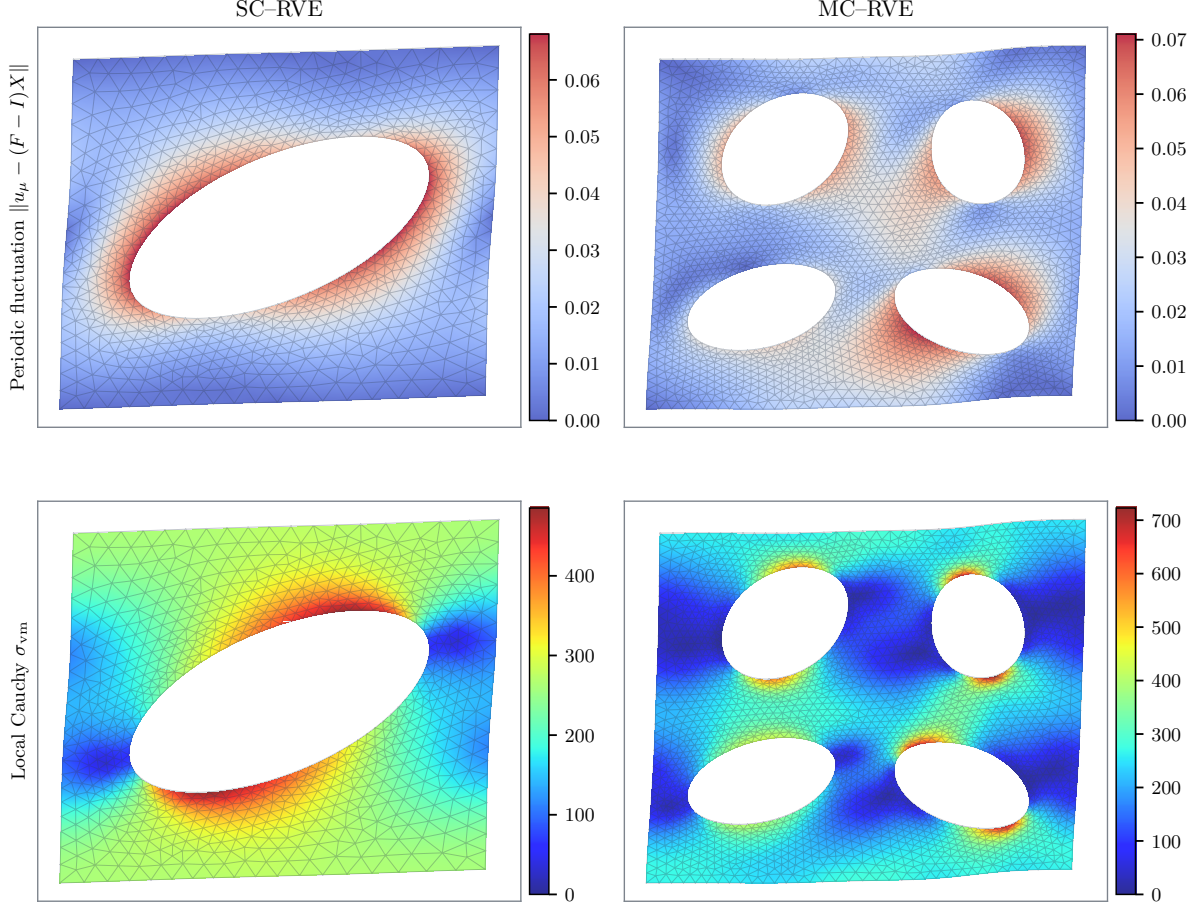


Figure 5: FOM microscopic fields at the common diagnostic state $(E_{11}, E_{22}, 2E_{12}) = (18, -4, 8)\%$, shown on the actual deformed cells without displacement amplification. Top: magnitude of the periodic displacement fluctuation after removing the imposed affine part. Bottom: local three-dimensional Cauchy von Mises stress, including the plane-strain out-of-plane stress; Gauss-point values are averaged to the nodes for visualization. Each panel uses its own color limits to resolve the spatial pattern within its respective cell; colors are therefore not a cross-cell magnitude comparison.

A common diagnostic state. The two cells respond differently to the same multiaxial deformation. Figure 5 applies $(E_{11}, E_{22}, 2E_{12}) = (18, -4, 8)\%$ to both FOMs. The inclined cavity of the SC-RVE concentrates the response around one dominant geometric direction, whereas the nonaligned cavities of the MC-RVE create several interacting localization regions. This geometric contrast motivates the distinct roles of the two benchmarks; the figure shows FOM fields only and is not an error test.

5.1.1 Training protocol

The two benchmarks answer different questions, but their energies are trained and selected in the same way, so that differences within each benchmark reflect the constitutive representation rather than the fitting procedure. Every model is a scalar energy $W(\mathbf{e}; \boldsymbol{\theta})$, whose first derivative gives the stress and whose second derivative gives the tangent $\mathbf{D}$ of Eq. (3). It is fitted to FOM energies and stresses at the fitting states and to the effective tangent $\mathbf{D}_0 = \mathbf{D}(\mathbf{0})$ of the periodic cell at the undeformed state.

The ICNN of Eq. (28) and the ICKAN of Eq. (29) have two hidden layers of widths 24 and 12, respectively, and each ICKAN connection combines six integrated-hat bases on fixed knots from -1.2 to 1.2 ($h = 0.48$). With six fixed paired features, the ICNN has 945 trainable parameters and the ICKAN 1629; learning the features adds only five parameters per feature. The initial features are selected from a finite bank of admissible candidates using only the fitting response and the reference tangent.³ Before entering either core,

³The bank contains 2028 candidates, which combine twelve orientations $\theta_i = k\pi/12$ with the stretch powers $p_i, q_i \in \{0.5, 1, 1.5, 2, 3\}$ and with determinant exponents at the bounds and midpoint of Eq. (25), moved slightly inside the admissible

each feature is centered at its undeformed value $z_i^0 = 1/p_i + 1/q_i$ and divided by its largest deviation from that value over the fitting states at initialization. The conditioned inputs therefore start in $[-1, 1]$, which the ICKAN knots cover with a small margin. The Unconstrained energy uses three softplus hidden layers of widths 128, 128, and 64 on the inputs of Eq. (19), for 25 473 parameters.

All trainable parameters, including the feature parameters of the learned variants, minimize

$$\mathcal{L}(\theta) = \frac{\sum_k \|s(e_k; \theta) - s_k\|_2^2}{\sum_k \|s_k\|_2^2} + 0.2 \frac{\sum_k [W(e_k; \theta) - W_k]^2}{\sum_k W_k^2} + 0.02 \frac{\|D(0; \theta) - D_0\|_F^2}{\|D_0\|_F^2}, \quad (65)$$

where the sums run over the fitting states e_k , with FOM energies W_k and stresses s_k . Stress carries the largest weight because it is queried throughout the FE² calculation. The energy term keeps the potential itself, and not only its gradient, close to the homogenized energy. The small tangent term anchors the stiffness at the undeformed state, where the vanishing stresses contribute little to the first term.

Optimization is full-batch and in double precision. The validation error is the first term of Eq. (65) evaluated on the validation states. Adam runs until this error stops improving, or for at most 200 000 steps, and L-BFGS then refines the best Adam state.⁴ The retained checkpoint is the one with the lowest validation error, and each model is trained from three seeds, 16, 29, and 47. Independent test states are evaluated only after every checkpoint has been frozen.

5.2 Multicavity RVE: learning the paired features

The paired-feature construction guarantees admissibility for every allowed parameter set, but it does not determine how much directional information the convex core needs. This question is particularly relevant for the MC-RVE, whose nonaligned cavities introduce several geometric directions. A small fixed feature set may then fail to distinguish deformation states with different directional responses. One remedy is to supply more features; another is to let their directions and exponents adapt to the RVE response. We examine both effects together by asking how accuracy changes with the number m of paired features and, for each m , what is gained by learning rather than fixing their parameters.

5.2.1 Cell, data, and controlled comparison

The MC-RVE distributes the void space among four unequal, off-center ellipses with different aspect ratios and orientations; see Fig. 6. Its several geometric directions make it the representation benchmark. The working mesh has 4621 quadratic triangles.

Refining the mesh to 8961 elements changes the homogenized energy, stress, and tangent by less than 0.1% over 64 audited states that include the corners of the strain domain used below; the sampled maximum microscopic first Piola–Kirchhoff stress norm changes by at most 4.06%. These comparisons support the working mesh for the representation study.

Controlled comparison. We sample $E_{11}, E_{22} \in [-0.04, 0.20]$ and $2E_{12} \in [-0.08, 0.08]$. The left panel of Fig. 7 shows the domain. The 4200 fitting states combine 4096 scrambled Sobol points in the interior with 96 points on the faces and the eight corners, so that the boundary of the box is sampled rather than reached by extrapolation. Independent Sobol sequences supply 512 validation and 512 independent-test states. Ten further straight paths run from the undeformed state to the boundary of the box in axial, biaxial, shear, and combined directions, each with 40 scored states; like the test states, they play no role in training or selection. For each $m \in \{2, 4, 6, 8, 16, 24, 32\}$, ICNN and ICKAN are trained in two variants that start from the same m initial features. The fixed variant holds their directions and exponents constant while training the convex core, whereas the learned variant optimizes those parameters subject to the constraints of Section 3. All other ingredients of Section 5.1.1 are shared within each core, so the difference between the two variants isolates the effect of adapting the features.

5.2.2 How many paired features are needed?

Figure 8 answers this question in the same way for both convex cores. Two paired features leave the response under-resolved, and both variants improve as m increases, but in different ways. The learned variants gain

interval so that the parameterization of Eq. (26) can represent them. Nonnegative least-squares fits of the fitting response and of the reference tangent identify the contributing candidates. The largest contributions are retained when more than m are selected, and pivoted QR completes the set when fewer are.

⁴Adam starts from learning rates of 5×10^{-3} for the ICNN and ICKAN and 4×10^{-4} for the Unconstrained energy, halved after 40 validation checks without improvement down to 10^{-5} , with a check every ten steps. It stops when the best validation error improves by less than 0.1% over 100 checks at the smallest learning rate. L-BFGS, in calls of at most ten iterations, stops under the same criterion over 30 calls.

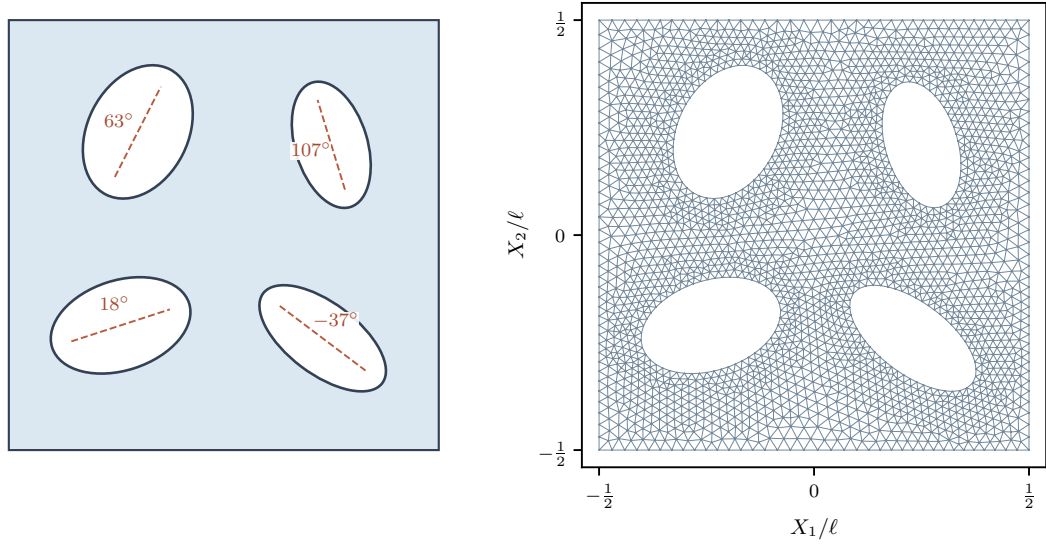


Figure 6: MC-RVE cavity geometry (left) and quadratic mesh (right). Dashed lines identify cavity major axes.

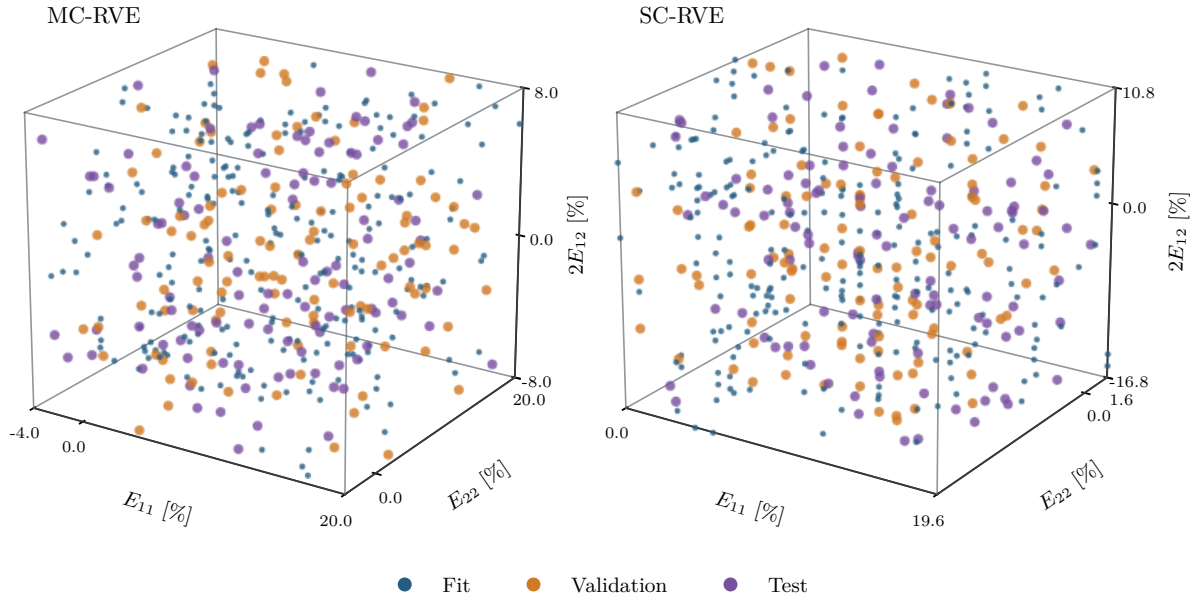


Figure 7: Constitutive sampling domains, in percent. Left: MC-RVE representation benchmark. Right: SC-RVE deployment benchmark. Wireframes show the prescribed limits; colors identify fitting, validation, and independent-test states.

most up to $m = 6$ and change by less than a factor of two beyond it, whereas the fixed variants continue to improve gradually up to $m = 32$. We therefore select $m = 6$, the first count at which both learned cores enter their low-error regime, for independent evaluation.

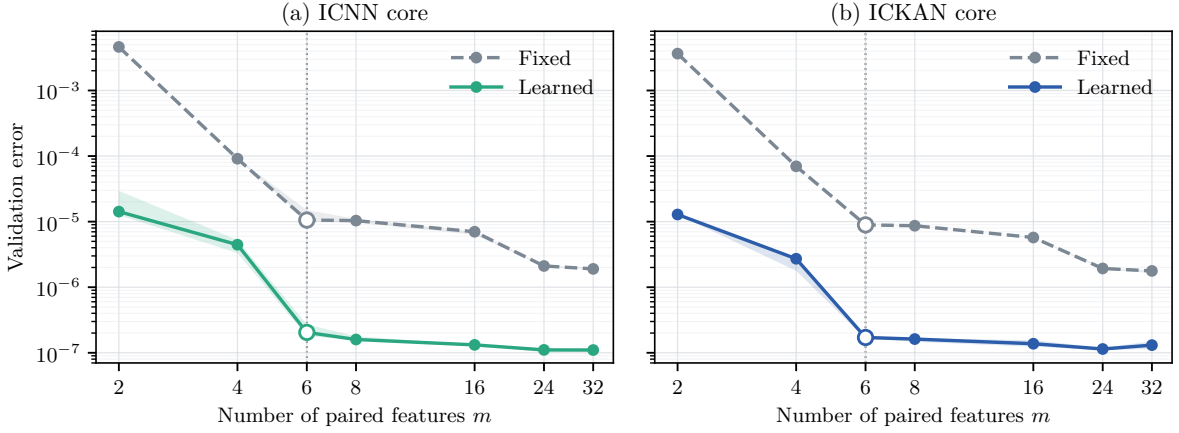


Figure 8: MC-RVE validation error, the relative squared stress error of Section 5.1.1, versus the number m of paired directional features. Lines show the median over three seeds and bands span their minimum–maximum range. The vertical guide marks $m = 6$, used for the independent comparison.

5.2.3 What does learning the features add?

Table 3 answers this question on the independent test states. At $m = 6$, learning the paired features lowers the ICNN median stress error from 0.328% to 0.0445% and the ICKAN error from 0.301% to 0.0406%, a reduction by a factor of approximately 7.4 in either core. The energy and the tangent, which is never fitted away from the undeformed state, improve by factors of about 14 and 5, respectively. Adding fixed directions does not close this gap. In the validation sweep, even 32 fixed features, with more than twice the trainable parameters of the learned six-feature models, remain about three times less accurate in relative stress error, a factor of about ten in the squared validation error of Fig. 8. The ten held-out paths confirm the ordering for stress, energy, and tangent on every path. The largest gain appears in pure shear, where the relative stress error falls from 3.2–4.0% with fixed features to 0.47–0.58% with learned ones, whereas biaxial compression produces the largest tangent errors for both variants. The similar gain for ICNN and ICKAN shows that the central improvement comes from adapting the directional representation rather than from one particular convex core.

The Unconstrained energy places these values in context. Without convexity or growth conditions, it reaches 0.0073% on the same states, about six times below either learned constrained model. Its network is also considerably larger. At matched capacity, the controls of Appendix F place the in-domain cost of the constraints on this cell at a factor of 3.2 to 5.5 in stress error; the one exception, a wide ICKAN, reflects early stopping rather than capacity.

Table 3: MC-RVE independent-test errors at $m = 6$. Each entry is the relative error norm of the stress $\mathbf{s}$, the energy W , or the tangent $\mathbf{D}$ of Eq. (3) over the 512 test states, as the median over three seeds. For every quantity, each learned seed is more accurate than each fixed seed. Adam stopped at a validation plateau for the fixed variants, whereas five of the six learned runs and the Unconstrained energy, which has no paired features, reached the 200 000-step cap.

Model	Stress [%]	Energy [%]	Tangent [%]
ICNN, fixed	0.3282	0.1096	1.9247
ICNN, learned	0.0445	0.0079	0.3851
ICKAN, fixed	0.3014	0.0943	1.7600
ICKAN, learned	0.0406	0.0066	0.3622
Unconstrained energy	0.0073	0.0011	0.0857

5.2.4 What do the constraints guarantee beyond the data?

For the ICNN and ICKAN energies, conditions (C1)–(C6) hold for every admissible parameter set by construction, whereas the Unconstrained energy is built to satisfy (C1)–(C4) only. Inside the box the difference does not show, since a bounded rank-one search over 520 states and 32 directions finds curvature above 236 MPa for all fifteen fits of Table 3. Farther out the two classes separate. Figure 9(a) follows equibiaxial compression, $\mathbf{F} = \sqrt{J} \mathbf{I}$, beyond the box. The minimum rank-one curvature of every Unconstrained fit changes sign between $J = 0.71$ and 0.73 , whereas the learned constrained fits remain above 220 MPa. A directed search over principal stretches between 0.55 and 1.5, in arbitrary orientations, shows that this loss is not tied to one path. It finds negative curvature for all three Unconstrained fits at more than a third of its 4096 sampled states, confirmed by energy finite differences, and only positive curvature for the twelve constrained fits, as their construction requires.

Figure 9(b) continues the same path toward volume collapse. The constrained energies diverge, as their logarithmic barrier guarantees, whereas the Unconstrained energy levels off at finite limits between 397 and 440 MPa. This contrast reflects the growth terms rather than polyconvexity, and an unconstrained network could receive the same barrier. No data constrain either energy at these states, so only the construction can guarantee (C5) and (C6) there.

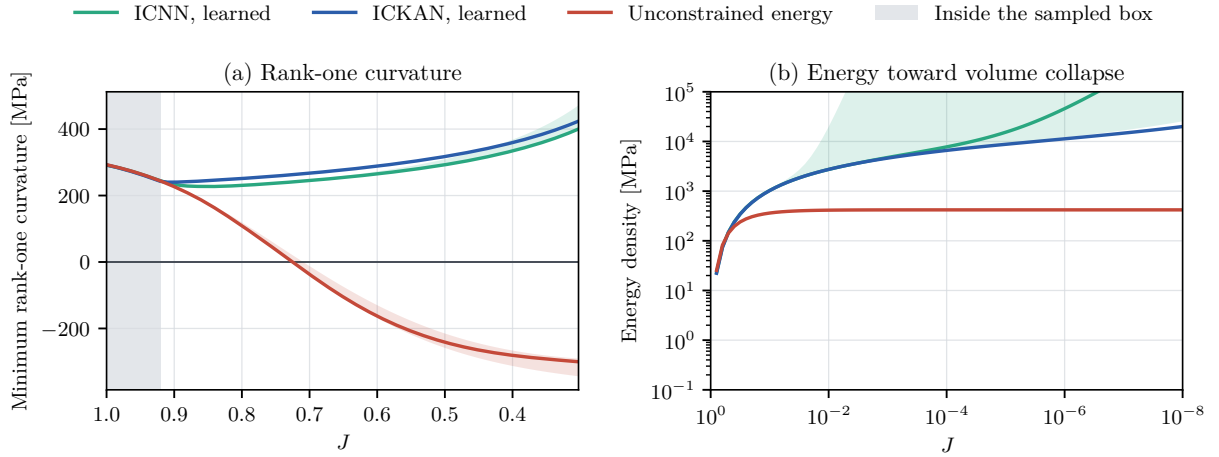


Figure 9: MC-RVE energies beyond the data along equibiaxial compression, $\mathbf{F} = \sqrt{J} \mathbf{I}$, for the learned ICNN and ICKAN energies and the Unconstrained energy. Lines show the median of three seeds and shaded bands their range. (a) Minimum rank-one curvature over unit directions. (b) Energy density toward volume collapse.

Global nonnegative energy, which the construction leaves to the saved parameters, is certified by the sufficient bound of Appendix C for all six learned fits, with interval margins between 8.5 and 10.8 in normalized energy units. For all fifteen fits, the reference energy and stress vanish to round-off, and centered differences reproduce the stress and the tangent to better than 10^{-9} of their largest entries.

Taken together, the MC-RVE supports the choices carried into the SC-RVE study. Six paired features give a compact representation, and adapting their directional parameters supplies the main accuracy gain for either convex core. The learned constrained energies then keep the median stress error below 0.05% while securing, beyond the data, the rank-one convexity and growth that the Unconstrained energy loses there.

5.3 Single-cavity RVE: constitutive accuracy and FE² deployment

5.3.1 Cell and coupon-derived data

The SC-RVE contains one centered elliptical void with aspect ratio 2 and major-axis angle 30° relative to the cell axes; see Fig. 10. Its working mesh has 1546 quadratic triangles and 6320 independent microscopic displacement coordinates.

At a uniaxial-stress state with $E_{11} = 0.20$, refining the mesh to 3095 elements changes the maximum microscopic first Piola–Kirchhoff stress norm by approximately 0.486% and the average axial stress by approximately 0.001%. This state lies on the path followed by the narrow section of the coupon in Section 5.3.5, beyond the largest axial strain reached there, so the check supports the mesh used for deployment.

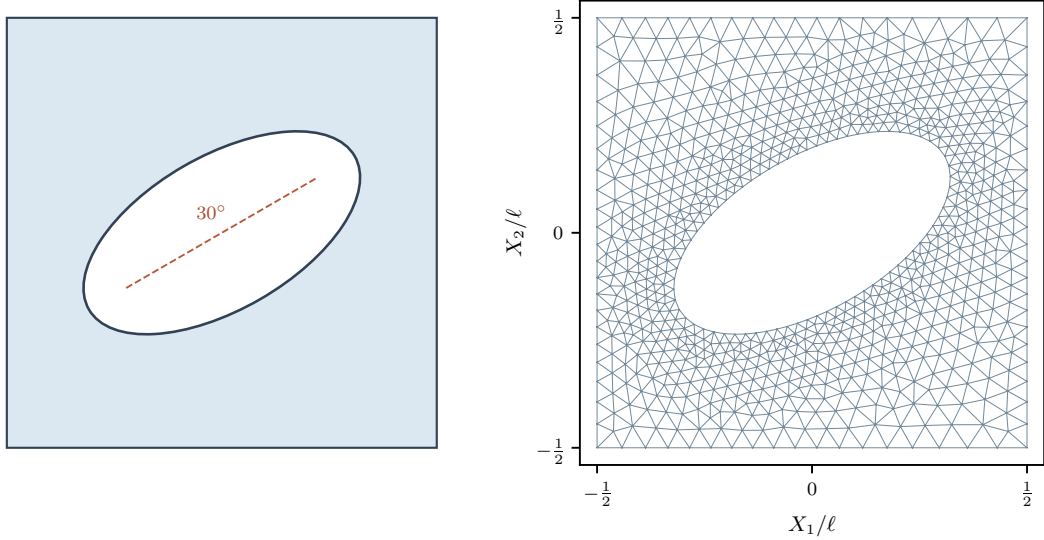


Figure 10: SC-RVE cavity geometry (left) and quadratic mesh (right). The dashed line identifies the cavity major axis.

Coupon-derived data. Unlike the MC-RVE, whose box is prescribed, the SC-RVE must supply the learned constitutive response throughout the coupon FE² calculation of Section 5.3.5, whose macroscopic quadrature points visit different strains during loading. We first ran a displacement-controlled coupon prepass with a St Venant–Kirchhoff approximation based on the cell reference tangent. The recorded strain envelope guides sampling only; RVE stresses provide the training labels. A second prepass with a softer response shifted the componentwise extrema by at most 5.1%, so we enlarged the spans by 40%. Because the coupon is tensile, the lower E_{11} bound is zero. In the engineering-strain coordinates $\mathbf{e} = (E_{11}, E_{22}, 2E_{12})^T$, the rounded bounds are

$$\mathbf{e}_{\min} = (0, -0.099, -0.168)^T, \quad \mathbf{e}_{\max} = (0.196, 0.016, 0.108)^T. \quad (66)$$

A Cartesian grid supplies 4950 nonreference states, of which 4208 are used for fitting and 742 for validation (split seed 5); an independent off-grid set of 400 states tests interpolation; see the right panel of Fig. 7. The same grid also provides neighboring states for the reduced microscopic models of Section 4.

5.3.2 Constitutive accuracy

The MC-RVE study identifies $m = 6$ as the first compact count at which both learned cores enter their low-error regime. We therefore ask whether that same count is sufficient for the SC-RVE deployment, without transferring feature parameters between the two geometries. Six initial features are selected anew from the SC-RVE fitting response and reference tangent, and the learned ICNN and ICKAN are trained as described in Section 5.1.1, as is the Unconstrained energy. For each model, the seed with the lowest validation error supplies the deployed model. Both constrained selections fall on seed 16, with the validation errors of the three seeds within 15% of one another, and the Unconstrained selection falls on seed 47.

Independent response. The 400 off-grid states now answer the deployment question directly. The compact ICNN and ICKAN attain aggregate stress errors of 0.621% and 0.636%, respectively, and the Unconstrained energy reaches 0.0052%, about 120 times lower; see Table 4. On this cell the construction therefore costs far more accuracy than on the MC-RVE, and capacity is not the reason. In the controls of Appendix F, cores widened to about 25 000 parameters change the constrained errors by less than 3%, whereas an Unconstrained energy with only 981 parameters remains about 75 to 80 times more accurate than either wide core. Thirty-two learned features lower the errors only to 0.537% and 0.618%. This behavior is consistent with the ordering argument of Section 3.3, since a monotone core cannot reverse the componentwise ordering of its features whatever its width; widening the core therefore cannot enlarge the response family beyond what the features and that ordering admit. The price of this restriction depends on the cell, a factor of 3.2 to 5.5 on the MC-RVE and of 76 to 126 on the SC-RVE at matched capacity. Comparable costs are reported for other constrained constructions. Linden et al. [14] found that imposing all common conditions on a small invariant-based network raised its stress error by about one order of magnitude for transverse isotropy, and still recommended the constrained model for its extrapolation. On homogenization data of a

matrix with a stiff spherical inclusion, the polyconvex models of Klein et al. [44] reached root-mean-square stress errors three to five times those of their non-polyconvex counterparts, and Kalina et al. [43] dropped polyconvexity after finding it too restrictive for their fits. The MC-RVE factors are of this size, whereas the SC-RVE shows that the price can be much larger for some microstructures.

Response beyond the box. A separate 350-state probe resolves the response outside the fitting box. It contains five increasing overshoot factors and labelled face, edge, and corner patterns; five states without a finite FOM label are excluded for every model, leaving 345. The aggregate probe errors rise to 10.6% and 10.1% for ICNN and ICKAN and to 0.668% for the Unconstrained energy, with the ringwise results in Supplementary Section S1. On this cell the constraints thus cost accuracy inside and outside the box. Inside it, where every coupon state lies, both constrained energies remain below 0.65%; what the constraints secure beyond it is examined next. Thirty-two features would also raise the probe errors to 35.6% and 16.9% and lose the nonnegative-energy certificate of Appendix C, so the coupon keeps the $m = 6$ models.

Table 4: SC-RVE independent-test constitutive errors of the selected models, in percent.

Model	e_S^{test} [%]	e_W^{test} [%]	Worst test [%]
ICNN	0.6214	0.1929	4.9263
ICKAN	0.6359	0.1900	4.8240
Unconstrained energy	0.0052	0.0010	0.0203

All values are percentages. Stress and energy errors are aggregate array norms; the last column is the maximum per-state stress error. SC-RVE: 400 independent test states.

5.3.3 Admissibility beyond the data

The comparison of Section 5.2.4 carries over to this cell; Supplementary Section S2 gives the sampled values. On the test, probe, and 12 000 additional in-box states, the rank-one curvature of all three energies stays above 135 MPa. Farther out, a radial expansion of the box to four times its size finds negative curvature for the Unconstrained energy at 14 108 of 109 743 states, first at 1.55 times the box and $J = 0.78$, and a broad scan of principal stretches between 0.25 and 3 finds it at 10 048 of 24 000 states. The ICNN and ICKAN remain above 176 MPa on both sets. The Unconstrained energy also stays finite along $\mathbf{F} = \sqrt{J} \mathbf{I}$ as $J \rightarrow 0^+$, because its network inputs remain bounded on that path, whereas the logarithmic barrier makes the constrained energies diverge. For the two deployed models, the sufficient bound of Appendix C certifies nonnegative energy, with interval margins of 4.85 and 4.91 in normalized energy units. For all three energies, derivative checks and the vanishing closed-cycle work of Supplementary Section S3 confirm the potential structure of the implementation.

5.3.4 Reduced models built from the same snapshots

The 4950 displacement snapshots provide the basis of the reduced microscopic models as well as the effective constitutive labels. Figure 11 shows how the relative truncation norm of their POD basis decays with the number of modes; for the retained 39-dimensional space it is 9.94×10^{-7} . After its orthogonal rotation into primary and secondary spaces, $n = 3$ and $\bar{n} = 36$ in the notation of Section 4.

The fitted relation between the deployed primary coordinates and macroscopic strain has an aggregate relative array discrepancy of 3.60×10^{-5} on these snapshots; its maximum absolute component discrepancy is 2.55×10^{-5} . These quantities explain why macroscopic strain is a useful primary-coordinate predictor for this material. The direct substitution in D-HPROM-ANN nevertheless remains an additional approximation.

The secondary-coordinate closure is a 3–64–64–36 feed-forward network with two tanh hidden layers and a linear output. Its training script uses input standardization, the raw secondary-coordinate loss in Eq. (51), and a 90/10 internal fit/validation split with seed 20260903. Full-batch Adam has initial learning rate 2×10^{-3} , a validation-driven reduction schedule, and a maximum budget of 20000 epochs. Its best relative validation mean-square error is 3.75×10^{-7} , an internal closure metric; the coupon of Section 5.3.5 provides the independent assessment.

Each deployed adaptive cubature field has architecture 3–128–128–128–10, GELU hidden activations, and the scaled softmax output of Eq. (58). The residual and stress checkpoints use distinct ten-element supports. Their larger neural cores illustrate why the online cost cannot be inferred solely from the three equilibrium coordinates or from the element count.

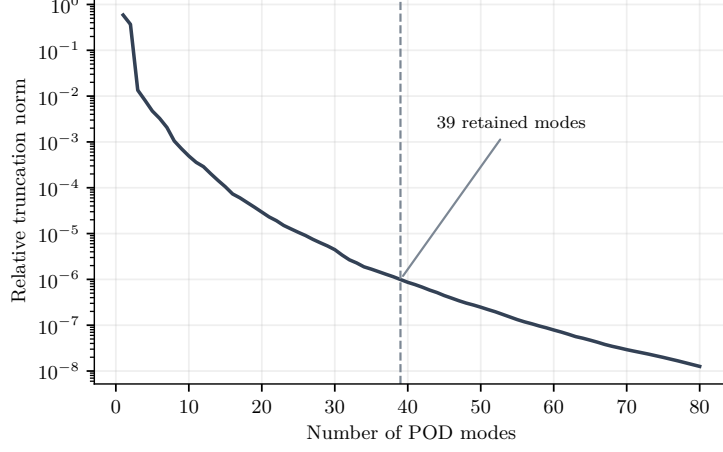


Figure 11: Relative truncation norm of the POD basis of the 4950 SC-RVE displacement snapshots versus the number of modes. The deployed basis retains 39 modes; its rotation into primary and secondary coordinates leaves this span unchanged.

Online models. The linear HPROM uses $n_{\text{tra}} = 39$ generalized coordinates, a residual rule supported on 135 elements, and a stress rule on 73 elements; their union contains 183 elements. The integrand-SVD tolerances are $\varepsilon_{\text{SVD}}^r = 10^{-3}$ and $\varepsilon_{\text{SVD}}^s = 10^{-5}$, retaining 134 and 72 integrand modes, respectively. The greedy integration tolerance in Eq. (56) is $\varepsilon_{\text{ECM}} = 10^{-6}$ for both rules, including the multiplier-sum constraint.

The iterative HPROM-ANN uses $n = 3$ primary and $\bar{n} = 36$ secondary coordinates, with ten residual and ten stress elements and a union of 19. Initial fixed-ECM candidates use $\varepsilon_{\text{SVD}} = 10^{-4}$ and $\varepsilon_{\text{ECM}} = 10^{-6}$; adaptive pruning targets ten elements per rule. This count is a prescribed pruning target, assessed through integration and response errors. The pruning uses up to 500 sampled states, tests up to 20 removal candidates per elimination, and applies only the local, positivity-enforcing stage of the algorithm; the residual weight field is then refitted on the same support with a longer training budget.

The direct model evaluates stress on ten elements and solves no microscopic equilibrium equations. All these counts refer to reduced computational meshes, not full-mesh evaluations followed by masking. Figure 12 shows their spatial distribution and the overlap of residual and output rules.

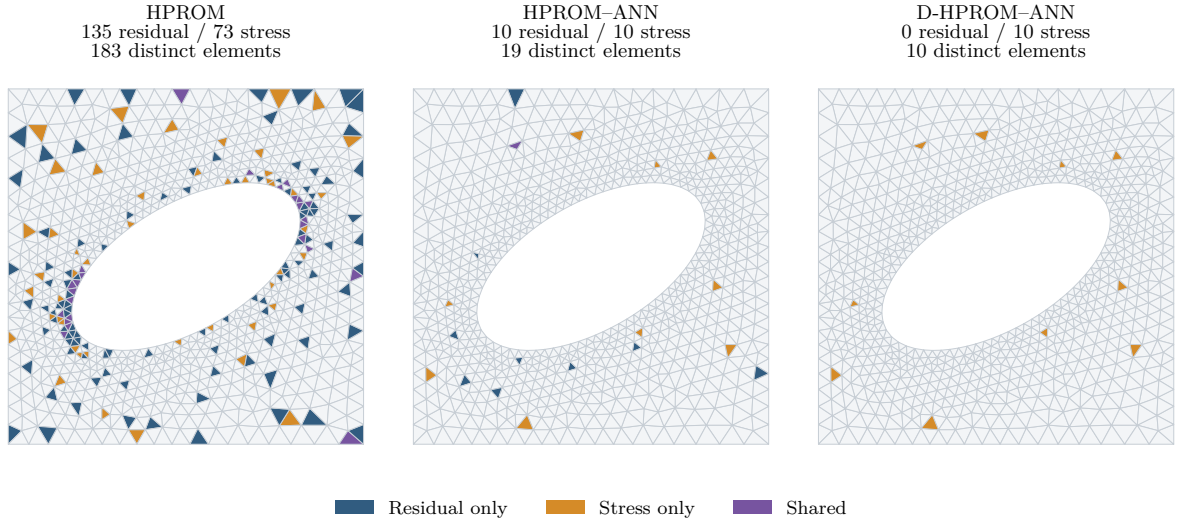


Figure 12: Element supports of the deployed hyperreduced microscopic models. The pale full-cell mesh provides location context only; discarded elements are not assembled online. Colors distinguish residual-only, stress-only, and shared elements.

5.3.5 Coupon FE²: accuracy and online cost

Purpose, boundary conditions, and coverage. We now use the SC-RVE to compare learned constitutive laws and reduced microscopic solvers within the same nonlinear structural calculation. The coupon measures how their approximations affect structural accuracy and online cost. The dog-bone profile has overall length 165 mm, grip width 19 mm, narrow-section width 13 mm, narrow-section length 57 mm, and fillet radius 76 mm. It is a plane-strain computational benchmark using an ASTM-inspired in-plane profile, not a simulation certified to reproduce an ASTM tensile test.

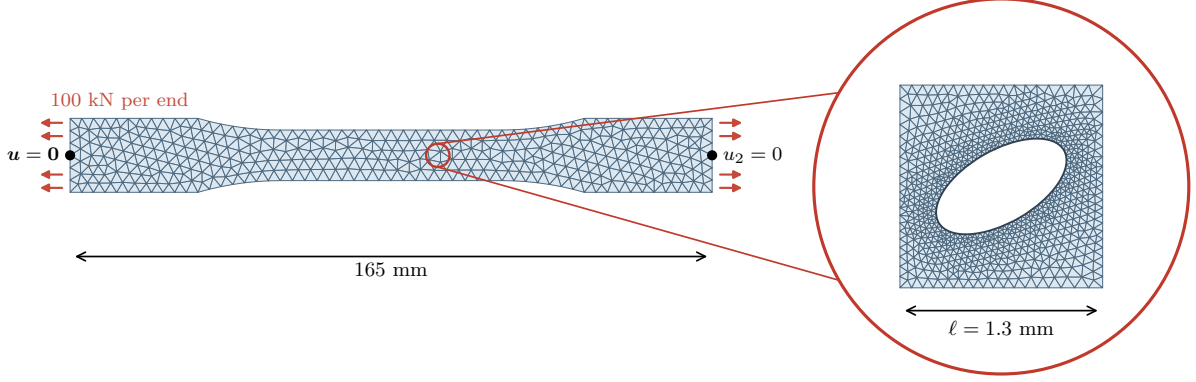


Figure 13: Coupon problem. Undeformed macro mesh with equal-and-opposite resultants of 100 kN per end; the three point constraints only remove rigid-body motion, since the two resultants balance each other. Every integration point queries the SC-RVE, solved by the FOM or replaced by a reduced model or a learned energy.

The two end faces receive equal-and-opposite axial resultants of 100 kN per end, applied in 20 equal increments (Fig. 13). Three point constraints remove rigid modes: the axial and transverse displacement at the left mid-edge node, and transverse displacement at the right mid-edge node. Entire end faces are not clamped. Consequently, a small skew of the deformed ends is not ruled out by the boundary conditions. The anisotropic response can couple axial extension to shear.

The macro mesh contains 690 quadratic triangles, 1501 nodes, and 2070 integration points. The reference extrusion thickness used to convert the two-dimensional stress integrals into the stated force resultants is 50 mm. This force convention is part of the plane-strain benchmark, not an ASTM specimen-thickness specification. The physical cell size is declared as 1.3 mm, corresponding to ten cells across the narrow width. This provides a scale-separation assumption, not a direct numerical validation of homogenization error at that ratio. All saved converged integration-point strains for every reported model remain inside the unchanged training box. Figure 14 shows the reference fields at the final load. The equivalent stress is nearly uniform in the narrow section, peaks slightly where the fillets begin, and decreases toward the grips.

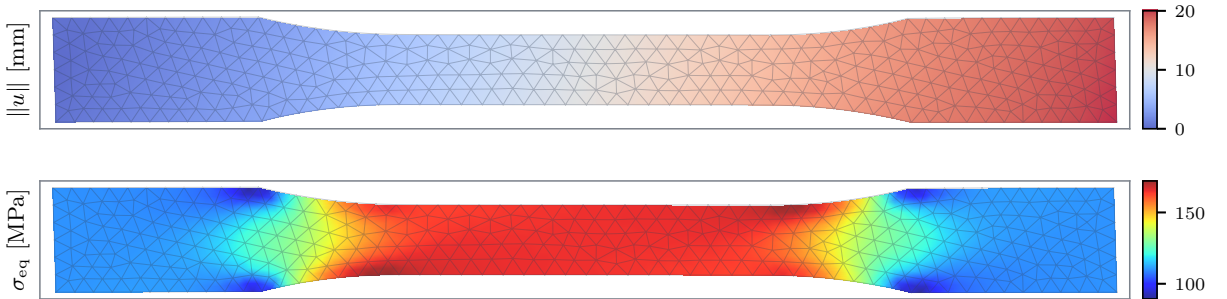


Figure 14: Nested FOM-FE² reference at 100 kN per end, shown on the deformed mesh without displacement amplification. Top: displacement magnitude. Bottom: in-plane von Mises equivalent of the macroscopic Cauchy stress, $\sigma_{eq} = (\sigma_{11}^2 - \sigma_{11}\sigma_{22} + \sigma_{22}^2 + 3\sigma_{12}^2)^{1/2}$; Gauss-point values are averaged to the nodes for visualization.

Error and timing definitions. For a saved nodal or integration-point array A , the reported relative error is

$$e_A = 100 \frac{\|\text{vec}(A - A_0)\|_2}{\|\text{vec}(A_0)\|_2}, \quad (67)$$

where A_0 is the nested reference on the same meshes. These are unweighted discrete array norms. The strain array stores engineering shear, whereas the stress array stores tensor shear. They are neither quadrature-weighted continuum L^2 norms nor invariant tensor Frobenius norms. The metric is kept identical across models; this does not make the three different field errors interchangeable.

Online wall time includes the complete macroscopic solve, microscopic or surrogate stress/tangent evaluations, communication, and lazy worker construction within the timed solve. Parent setup, offline data generation, training, reduced-mesh construction outside the timed region, and result export are excluded. The FOM uses 20 worker processes with one numerical-library thread per worker. The three optimized microscopic surrogates use the same process configuration and three sequential fresh complete runs. PANN evaluations are batched in one process, with the number of PyTorch threads selected by a preliminary sweep.

The macroscopic solve stops when the free-degree-of-freedom residual norm falls below 10^{-5} N or its ratio to the initial increment residual falls below 10^{-7} . The periodic FOM uses a relative microscopic displacement-increment tolerance of 10^{-10} , with continuation when needed. The FOM and PANN times come from single runs, so their ratios are observed online speed-ups under the stated implementations rather than matched-thread efficiency estimates.

The affine HPROM uses its exact reduced residual Jacobian, whereas HPROM-ANN uses the projected-stiffness modified-Newton matrix described in Section 4.4.2; both use a microscopic increment tolerance of 10^{-10} . Their effective constitutive tangents nevertheless account for the equilibrium sensitivity of their respective full residuals. The reported speed-up compares these implemented workflows, including their different iteration matrices and derivative costs; it does not isolate a nonlinear-representation gain or quantify the iteration penalty of modified Newton.

Table 5: Coupon simulations: final-field errors of Eq. (67) with respect to the nested FOM, online wall time, and speed-up over the FOM.

Model	e_u [%]	e_E [%]	e_S [%]	Time [s]	Speed-up
FOM	—	—	—	4268.68	—
HPROM	0.0057	0.0165	0.0010	67.35	63.38
HPROM-ANN	0.0520	0.1497	0.0373	30.14	141.61
D-HPROM-ANN	0.0232	0.0485	0.0097	3.79	1126.90
ICNN	0.4066	0.6785	0.0739	1.04	4105.14
ICKAN	0.4197	0.6914	0.0732	1.95	2187.66
Unconstrained energy	0.0022	0.0050	0.0006	1.48	2884.31

Accuracy–cost trade-off and Newton behavior. Table 5 shows distinct operating points rather than a single universally best model. Among the reduced microscopic models, the linear HPROM gives the smallest displacement error, approximately 0.0057%, at a median online time of 67.35 s. The iterative HPROM-ANN reduces this time to 30.14 s with a displacement error of approximately 0.0520%. The iterative nonlinear model is therefore about 2.23 times faster than the linear one, both in their optimized implementations. The direct closure reduces the time to 3.79 s and has a displacement error of approximately 0.0232% on this path. Its lower error than the iterative nonlinear model is an observed outcome, not a consequence of a general error ordering.

The six-feature ICNN and ICKAN take 1.04 s and 1.95 s, with displacement errors of approximately 0.407% and 0.420%, below their constitutive test errors but above those of the reduced microscopic models on this benchmark. The Unconstrained energy takes 1.48 s and gives the smallest displacement error in Table 5, 0.0022%, which mirrors the constitutive ordering of Section 5.3.2. Every saved coupon state lies inside the box, where all three energies have positive rank-one curvature at every sampled state, so this tensile path does not exercise the guarantees that separate them. On this path the constrained energies trade about two orders of magnitude in structural accuracy for admissibility beyond the data, while keeping the displacement error near 0.4%.

All models require four macroscopic iterations per increment, so this example provides no evidence that polyconvexity reduces Newton iterations. Inside the box, the constrained energies thus run in the common structural solver like the Unconstrained energy, at comparable cost. Their guarantees matter at states that the coupon never reaches, and the next example reaches them deliberately.

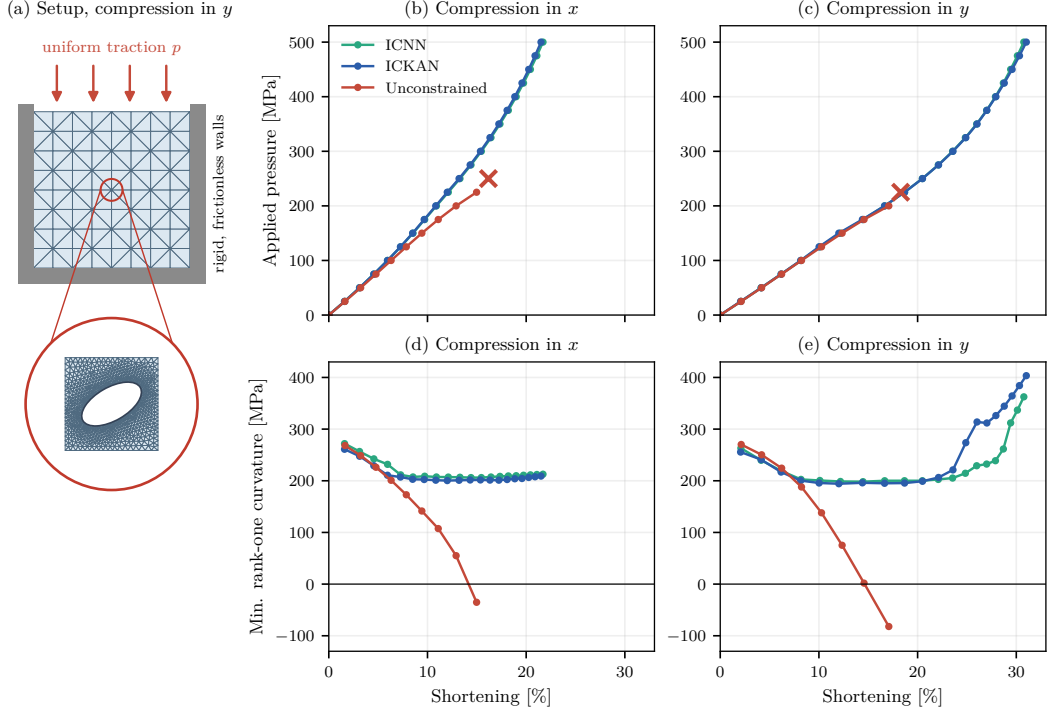


Figure 15: Confined compression beyond the data with the SC-RVE energies. (a) Setup for compression in y : the homogenized block, meshed with 128 quadratic triangles, evaluates the SC-RVE energy at every integration point; for compression in x the traction acts on a vertical face. (b, c) Applied pressure versus shortening, one marker per increment of 25 MPa; a cross marks the increment in which Newton does not converge. (d, e) Minimum rank-one curvature over the 384 Gauss points of the block at each converged increment.

5.3.6 Confined compression beyond the data

Confined compression reaches such states through compaction. A square block of side 10 mm, meshed with 128 quadratic triangles, is compressed by a uniform traction on one face while rigid, frictionless walls fix the normal displacement of the other three (Fig. 15a). The three SC-RVE energies, the assembly, and the Newton iteration are those of the coupon, and the load is applied in 20 equal increments up to 500 MPa, in x or in y . Compression in x leaves the fitting box with the first increment, since the box contains no axial compression, and compression in y leaves it beyond about 10% shortening. At every converged increment we also evaluate the rank-one curvature of Eq. (68) at the 384 Gauss points of the block, minimized over unit directions. Where it is positive, the incremental equilibrium problem is strongly elliptic.

Both constrained energies reach 500 MPa in both directions, with four to nine Newton iterations per increment and final shortenings of about 22% in x and 31% in y (Fig. 15b,c). The Unconstrained energy follows a softer curve in x and nearly the same curve in y , until Newton does not converge within 50 iterations in the increment to 250 MPa in x and to 225 MPa in y . This stop coincides with a loss of rank-one convexity in the block (Fig. 15d,e). The minimum curvature of the Unconstrained energy falls steadily with the load and turns negative at the last converged increment, at one of the 384 Gauss points in x and at two in y . These points lie in the elements at the corner where the left and bottom walls meet, where the local volume ratio has fallen to J between 0.61 and 0.72. The constrained energies, rank-one convex by construction, keep the minimum curvature above 194 MPa at every Gauss point and increment. The test does not compare accuracy, because neither data nor a reference exist at these states; it shows which energies keep the structural problem solvable once it leaves the data.

6 Conclusions

Paired directional features with constrained trainable powers and determinant exponents provide a practical polyconvex constitutive construction for the anisotropic periodic cells studied here. Their isotropic reference derivatives allow exact stress normalization through an affine determinant correction without tying the normalization coefficient to the volumetric barrier stiffness. The proof combines feature convexity in

independent minors with core convexity and monotonicity, and includes every correction in the complete energy. Global nonnegative energy is supported separately by a sufficient bound evaluated for the saved parameters, which certifies every learned six-feature model of this study.

The two cells separate representation from deployment. On the multicavity cell, learning six paired features reduces the held-out stress error sevenfold, to about 0.04%, and prescribing more directions does not close this gap. On the single-cavity cell, the selected ICNN and ICKAN attain approximately 0.63% held-out stress error and below 0.42% displacement error in the coupon simulation. The Unconstrained energy, trained under the same protocol, is more accurate inside the data by a factor of about six on the first cell and about 120 on the second, and controls of capacity and feature count attribute this price to the admissible construction rather than to network size. Beyond the data, the Unconstrained energy loses rank-one convexity and remains finite under volume collapse, whereas the constrained energies are rank-one convex everywhere and grow without bound as the volume vanishes.

Optimized reduced microscopic models provide a complementary accuracy–cost reference. They are more accurate than the constrained energies on the coupon, while direct learned constitutive evaluation is generally faster, and all compared models converge with the same number of macroscopic iterations per increment. Because every coupon state lies inside the data, that example does not exercise the guarantees, and the Unconstrained energy is the most accurate model there. Confined compression takes the same energies beyond the data. The constrained energies then carry the block to 500 MPa in both directions, whereas with the Unconstrained energy Newton stops below half that load, exactly when its rank-one curvature turns negative at a corner of the block. The accuracy given up is thus the price of a structural problem that remains solvable beyond the data, which is the property the construction is designed to secure.

A Rank-one paths and finite-strain energy curvature

This appendix supplies the algebra behind the two finite-strain curvature statements in criterion (C5). For a twice differentiable energy, nonnegative curvature along every rank-one perturbation is expressed by the Legendre–Hadamard condition

$$\partial_{\mathbf{F}\mathbf{F}}^2 W(\mathbf{F})[\mathbf{a} \otimes \mathbf{b}, \mathbf{a} \otimes \mathbf{b}] \geq 0 \quad \text{for all } \mathbf{a}, \mathbf{b} \in \mathbb{R}^2. \quad (68)$$

The polyconvex representation in Eq. (18) implies this condition because the determinant is affine along rank-one paths. For 2×2 matrices,

$$\det(\mathbf{F} + t\mathbf{H}) = \det \mathbf{F} + t \operatorname{cof} \mathbf{F} : \mathbf{H} + t^2 \det \mathbf{H}, \quad \operatorname{cof} \mathbf{F} = \begin{pmatrix} F_{22} & -F_{21} \\ -F_{12} & F_{11} \end{pmatrix}, \quad (69)$$

and $\det(\mathbf{a} \otimes \mathbf{b}) = 0$. On a segment $\mathbf{F}_s = (1-s)\mathbf{F}_A + s\mathbf{F}_B$ with $\mathbf{F}_B - \mathbf{F}_A = \mathbf{a} \otimes \mathbf{b}$, the pair $(\mathbf{F}_s, \det \mathbf{F}_s)$ is therefore the corresponding convex combination of its endpoint values, and it stays admissible when both endpoint determinants are positive. Convexity of $\mathcal{P}$ then gives $W(\mathbf{F}_s) \leq (1-s)W(\mathbf{F}_A) + sW(\mathbf{F}_B)$, so a twice differentiable W has nonnegative second derivative along the segment, which is Eq. (68) [17, 18].

For an energy expressed in strain, the curvature along $\mathbf{F} + t\mathbf{H}$ follows from the chain rule. Let $\mathbf{H} \in \mathbb{R}^{2 \times 2}$ be an arbitrary perturbation and let t parametrize $\mathbf{F}(t) = \mathbf{F} + t\mathbf{H}$. Dots denote derivatives with respect to t , and $\operatorname{sym}(\mathbf{A}) = (\mathbf{A} + \mathbf{A}^T)/2$ denotes the symmetric part of a matrix. Direct multiplication gives

$$\begin{aligned} \mathbf{E}(t) &= \frac{1}{2} [\mathbf{F}^T \mathbf{F} - \mathbf{I} + t(\mathbf{F}^T \mathbf{H} + \mathbf{H}^T \mathbf{F}) + t^2 \mathbf{H}^T \mathbf{H}] \\ &= \mathbf{E}(0) + t \operatorname{sym}(\mathbf{F}^T \mathbf{H}) + \frac{1}{2} t^2 \mathbf{H}^T \mathbf{H}. \end{aligned} \quad (70)$$

It follows that $\dot{\mathbf{E}}(0) = \operatorname{sym}(\mathbf{F}^T \mathbf{H})$ and $\ddot{\mathbf{E}}(0) = \mathbf{H}^T \mathbf{H}$. Using the engineering-strain convention of Eq. (2), the chain and product rules give

$$\begin{aligned} \frac{dW}{dt} &= \mathbf{s} \cdot \dot{\mathbf{e}}, \\ \frac{d^2 W}{dt^2} &= \sum_{i,j} \frac{\partial s_i}{\partial e_j} \dot{e}_j \dot{e}_i + \sum_i s_i \ddot{e}_i = \dot{\mathbf{e}}^T \mathbf{D} \dot{\mathbf{e}} + \mathbf{S} : \ddot{\mathbf{E}}. \end{aligned} \quad (71)$$

Substituting $\ddot{\mathbf{E}}(0)$ gives

$$\left. \frac{d^2 W(\mathbf{F} + t\mathbf{H})}{dt^2} \right|_{t=0} = \dot{\mathbf{e}}(0)^T \mathbf{D} \dot{\mathbf{e}}(0) + \mathbf{S} : (\mathbf{H}^T \mathbf{H}). \quad (72)$$

Here $\mathbf{D} = \partial \mathbf{s} / \partial \mathbf{e}$ is the constitutive tangent and $\mathbf{S}$ is the second Piola–Kirchhoff stress, both evaluated at $t = 0$. The first term accounts for the change of stress with strain, and the second for the curvature of the

strain path, weighted by the current stress. Even if $\mathbf{D}$ is positive semidefinite, the stress term can be negative in compression.

A simple strain-convex energy shows that the stress term can make this curvature negative. Consider $W(\mathbf{E}) = \mu \mathbf{E} : \mathbf{E}$, where $\mu > 0$ is a constant with units of energy density. One has $\mathbf{S} = 2\mu \mathbf{E}$. Since

$$W(\mathbf{e}) = \mu(e_1^2 + e_2^2 + e_3^2/2), \quad \mathbf{D} = \text{diag}(2\mu, 2\mu, \mu), \quad (73)$$

the energy is strictly convex in engineering strain. At $\mathbf{F} = \lambda \mathbf{I}$, where $\lambda > 0$ is the common in-plane stretch, choose the rank-one perturbation $\mathbf{H} = \text{diag}(1, 0)$, which keeps $\det(\lambda \mathbf{I} + t \mathbf{H}) = \lambda(\lambda + t)$ positive for $t > -\lambda$. Then

$$\dot{\mathbf{e}} = (\lambda, 0, 0)^T, \quad \mathbf{S} = \mu(\lambda^2 - 1)\mathbf{I}, \quad \mathbf{H}^T \mathbf{H} = \text{diag}(1, 0). \quad (74)$$

Hence the tangent term is $2\mu\lambda^2$, and the stress term is $\mu(\lambda^2 - 1)$, giving

$$\left. \frac{d^2 W(\lambda \mathbf{I} + t \mathbf{H})}{dt^2} \right|_{t=0} = 2\mu\lambda^2 + \mu(\lambda^2 - 1) = \mu(3\lambda^2 - 1). \quad (75)$$

At $\lambda = 1/2$, the curvature is $-\mu/4$, violating Eq. (68) despite convexity in $\mathbf{E}$. The curvature is negative only for $\lambda < 1/\sqrt{3}$, so the counterexample requires sufficiently strong compression.

B Proofs for the paired features and complete energy

This appendix proves Propositions 1 and 2, starting from a single stretch-determinant contribution.

B.1 Joint convexity of the directional building block

Every contribution in Eq. (24) has the form

$$f(\mathbf{x}, J) = \|\mathbf{x}\|^a J^{-b}, \quad \mathbf{x} = \mathbf{F} \mathbf{d}, \quad (76)$$

up to a positive constant, with $a = 2p$ or $2q$. It suffices to determine when this building block is jointly convex in $(\mathbf{x}, J)$. Let $t = \|\mathbf{x}\| > 0$. Because f depends on $\mathbf{x}$ only through t , its Hessian separates into a block coupling the radial variable t to J and transverse directions orthogonal to $\mathbf{x}$. The radial-determinant block is

$$\begin{pmatrix} a(a-1)t^{a-2}J^{-b} & -abt^{a-1}J^{-b-1} \\ -abt^{a-1}J^{-b-1} & b(b+1)t^aJ^{-b-2} \end{pmatrix}. \quad (77)$$

For this symmetric 2×2 block to be positive semidefinite, its diagonal entries and determinant must be nonnegative. The diagonal entries are nonnegative for $a \geq 1$ and $b \geq 0$, while the determinant is

$$\begin{aligned} \det &= ab[(a-1)(b+1) - ab]t^{2a-2}J^{-2b-2} \\ &= ab(a-b-1)t^{2a-2}J^{-2b-2} \geq 0. \end{aligned} \quad (78)$$

The factor $a-b-1$ gives the upper bound $b \leq a-1$. If it is violated, the determinant is negative and the block has curvatures of opposite sign. The remaining, transverse Hessian eigenvalues are $at^{a-2}J^{-b} \geq 0$. Hence $a \geq 1$ and $0 \leq b \leq a-1$ make the complete Hessian positive semidefinite. Continuous extension gives convexity at $\mathbf{x} = \mathbf{0}$. That point is never reached on $\text{GL}^+(2)$, because $\mathbf{F} \mathbf{d} \neq \mathbf{0}$ for a nonzero direction, so the lack of smoothness of $\|\mathbf{x}\|^a$ there does not affect the stress or tangent. Substituting $a = 2p$ gives $p \geq \frac{1}{2}$ and $0 \leq b \leq 2p-1$, and the same argument with (q, c) establishes all bounds in Eq. (25).

B.2 Proof of Proposition 1

Proof. Convexity and objectivity. For any fixed learned parameter set, the directions $\mathbf{d}_i$ and $\mathbf{d}_i^\perp$ are independent of deformation, so the maps $\mathbf{F} \mapsto \mathbf{F} \mathbf{d}_i$ and $\mathbf{F} \mapsto \mathbf{F} \mathbf{d}_i^\perp$ are linear. Composing the convex building block above with these maps, multiplying by the positive factors $1/p_i$ and $1/q_i$, and adding the two contributions proves convexity of z_i in independent $(\mathbf{F}, J)$. A left proper rotation changes $\mathbf{F} \mathbf{d}$ to $\mathbf{Q} \mathbf{F} \mathbf{d}$ without changing its norm, so each contribution is objective.

Reference value and derivative. The physical relation $J = \det \mathbf{F}$ is now restored, with every learned parameter still fixed. At $\mathbf{E} = \mathbf{0}$, both directional invariants and J equal one, so the two contributions give $z_i^0 = 1/p_i + 1/q_i$. Moreover, $I_4(\mathbf{d}_i) = \mathbf{d}_i^T (\mathbf{I} + 2\mathbf{E}) \mathbf{d}_i$ and $J^2 = (1 + 2E_{11})(1 + 2E_{22}) - 4E_{12}^2$ imply $\partial_{\mathbf{E}} I_4(\mathbf{d}_i) = 2\mathbf{d}_i \otimes \mathbf{d}_i$ and $\partial_{\mathbf{E}} J|_0 = \mathbf{I}$. Applying the product rule to the first contribution gives

$$\partial_{\mathbf{E}} \left. \frac{I_4(\mathbf{d}_i)^{p_i}}{p_i J^{b_i}} \right|_0 = \frac{1}{p_i} [2p_i \mathbf{d}_i \otimes \mathbf{d}_i - b_i \mathbf{I}] = 2\mathbf{d}_i \otimes \mathbf{d}_i - \frac{b_i}{p_i} \mathbf{I}. \quad (79)$$

The perpendicular contribution has the same form with $(\mathbf{d}_i, p_i, b_i)$ replaced by $(\mathbf{d}_i^\perp, q_i, c_i)$. Since $\mathbf{d}_i \otimes \mathbf{d}_i + \mathbf{d}_i^\perp \otimes \mathbf{d}_i^\perp = \mathbf{I}$, adding the two contributions removes the directional dependence at the reference and gives

$$\partial_{\mathbf{E}} z_i|_0 = 2(\mathbf{d}_i \otimes \mathbf{d}_i + \mathbf{d}_i^\perp \otimes \mathbf{d}_i^\perp) - \left(\frac{b_i}{p_i} + \frac{c_i}{q_i} \right) \mathbf{I} = \rho_i \mathbf{I}. \quad (80)$$

In the work-conjugate engineering-strain convention of Eq. (2), this tensor gradient corresponds to $\partial_{\mathbf{e}} z_i|_0 = \rho_i(1, 1, 0)^T$, proving Eq. (27). $\square$

B.3 Proof of Proposition 2

Proof. Potential structure, objectivity, and stress symmetry (C1)–(C3). Every directional norm, J , and $\text{tr } \mathbf{C}$ are unchanged by left proper rotations, since all of them can be expressed through $\mathbf{d}_i^T \mathbf{C} \mathbf{d}_i$, $\text{tr } \mathbf{C}$, and $\sqrt{\det \mathbf{C}}$. Hence the complete energy is objective, and differentiation with respect to symmetric strain gives a symmetric material stress. Differentiating this scalar energy supplies (C1) and the major symmetry of its constitutive tangent. Smoothness holds on $\text{GL}^+(2)$ because $J > 0$, nonzero material directions have nonzero deformed length, and both cores are C^2 . Because this argument uses left rotations only, it does not impose a material symmetry group.

Reference normalization (C4). Equation (37) shows that W_{learn} sets the reference energy, and Eq. (39) shows that W_{corr} cancels the learned reference stress. The calculations for W_{vol} and W_{dist} in Section 3.5 show that both growth terms have zero value and zero strain gradient at the reference, so the complete energy satisfies both reference conditions.

Polyconvexity (C5). Let $\mathbf{G} \in \mathbb{R}^{2 \times 2}$ and $j > 0$ be independent. Define $\mathcal{P}(\mathbf{G}, j)$ by the right-hand side of Eq. (36), replacing $\mathbf{z}$ by $\mathbf{z}(\mathbf{G}, j)$, J by j , and $\text{tr } \mathbf{C}$ by $\|\mathbf{G}\|_F^2$. For two such pairs $\mathbf{X}_1 = (\mathbf{G}_1, j_1)$, $\mathbf{X}_2 = (\mathbf{G}_2, j_2)$ and $0 \leq t \leq 1$,

$$\begin{aligned} H(\mathbf{z}(t\mathbf{X}_1 + (1-t)\mathbf{X}_2)) &\leq H(t\mathbf{z}(\mathbf{X}_1) + (1-t)\mathbf{z}(\mathbf{X}_2)) \\ &\leq tH(\mathbf{z}(\mathbf{X}_1)) + (1-t)H(\mathbf{z}(\mathbf{X}_2)). \end{aligned} \quad (81)$$

The first inequality uses feature convexity and core monotonicity, and the second uses core convexity, so W_{learn} is convex up to its constant reference subtraction. The correction $W_{\text{corr}} = -r(j-1)$ is affine for either sign of r , while W_{vol} and W_{dist} are convex in independent $(\mathbf{G}, j)$ because their nonaffine parts are positive multiples of $-\log j$, $(j-1)^2$, and $\|\mathbf{G}\|_F^2$. Consequently, $\mathcal{P}$ is convex and $W(\mathbf{F}) = \mathcal{P}(\mathbf{F}, \det \mathbf{F})$, establishing Eq. (18).

Growth (C6). Convexity of H gives its supporting-plane inequality at $\mathbf{z}^0$,

$$H(\mathbf{z}) \geq H(\mathbf{z}^0) + \sum_i h_i(z_i - z_i^0) \geq H(\mathbf{z}^0) - \sum_i h_i z_i^0. \quad (82)$$

The second inequality uses $h_i, z_i \geq 0$. Hence W_{learn} has a deformation-independent lower bound and cannot cancel the analytic growth terms. As $J \rightarrow 0^+$, the logarithmic part of $W_{\text{vol}} + W_{\text{dist}}$ diverges while W_{corr} remains bounded. As $J \rightarrow \infty$, the positive quadratic part of W_{vol} dominates every potentially negative linear or logarithmic term. Finally, if J remains in a compact subset of $(0, \infty)$ while $\|\mathbf{F}\|_F \rightarrow \infty$, the term $\varepsilon \|\mathbf{F}\|_F^2/2$ in W_{dist} diverges and all determinant-only terms remain bounded. These are the three stated growth limits. $\square$

C Sufficient lower bound for nonnegative energy

Define $k_i = 1/p_i + 1/q_i$ and $\eta_i = \rho_i/k_i$. Write $t = \|\mathbf{F} \mathbf{d}_i\|^2$, $u = \|\mathbf{F} \mathbf{d}_i^\perp\|^2$. Hadamard's determinant inequality gives $tu \geq J^2$. Since each feature is increasing in t, u , its minimum at fixed J occurs on $tu = J^2$. Minimizing

$$\frac{t^{p_i}}{p_i J^{b_i}} + \frac{J^{2q_i - c_i} t^{-q_i}}{q_i} \quad (83)$$

gives $t^{p_i+q_i} = J^{2q_i+b_i-c_i}$ and hence $z_i \geq k_i J^{\eta_i}$. The bound holds separately for each feature, without assuming that the minima for different directions are attained simultaneously, and $h_i \geq 0$ allows it to be inserted term by term in Eq. (82).

Inserting the feature bound and using $\text{tr } \mathbf{C}/2 \geq J$ in W_{dist} gives

$$W \geq g(J) := \sum_i h_i k_i [J^{\eta_i} - 1 - \eta_i(J-1)] + (\alpha + \varepsilon)(J-1 - \log J) + \frac{\beta}{2}(J-1)^2. \quad (84)$$

Because $g(1) = g'(1) = 0$, it remains to establish $g''(J) \geq 0$ for every $J > 0$. Differentiation gives

$$J^2 g''(J) = \alpha + \varepsilon + \beta J^2 + \sum_i c_i^* J^{\eta_i}, \quad c_i^* = h_i k_i \eta_i (\eta_i - 1). \quad (85)$$

Only terms with $0 < \eta_i < 1$ can have negative coefficients. If $c_- = \sum_{c_i^* < 0} |c_i^*|$, the two inequalities

$$\alpha + \varepsilon + \sum_{\eta_i < 0} c_i^* - c_- \geq 0, \quad (86)$$

$$\beta + \sum_{\eta_i \geq 1} c_i^* - c_- \geq 0 \quad (87)$$

are sufficient on $0 < J \leq 1$ and $J \geq 1$, respectively. For the first region, $J^{\eta_i} \leq 1$ for negative-coefficient terms and $J^{\eta_i} \geq 1$ when $\eta_i < 0$. For the second region, dividing $J^2 g''$ by J makes every negative-coefficient power satisfy $J^{\eta_i-1} \leq 1$, while $J^{\eta_i-1} \geq 1$ for $\eta_i \geq 1$ and $\beta J \geq \beta$. Dropping the remaining nonnegative contributions gives the stated bounds. Their failure is inconclusive, so the trained parameters are checked with a sharper interval bound.

The audit partitions $[10^{-4}, 10^4]$ into 4000 logarithmically spaced intervals. On each interval $[J_L, J_R]$, a positive power term is bounded below by the smaller endpoint value, whereas a negative-coefficient term is bounded below by $c_i^* J_R^{\eta_i}$, since its exponent lies in $(0, 1)$. The constant and quadratic terms are bounded below by $\alpha + \varepsilon$ and βJ_L^2 . Below the finite interval, the nonnegative negative-exponent terms increase and the magnitudes of the negative-coefficient terms decrease toward $J = 0$. Above it, factoring out J^{η_-} with $\eta_- = \max_{c_i^* < 0} \eta_i < 1$ yields a lower bound whose retained positive powers increase. These endpoint and tail inequalities cover every $J > 0$, not only the grid points. Positive margins establish the sufficient bound for the trained parameters, up to floating-point accuracy, which random energy samples could not do.

D Latent-space closure details

This appendix supplies the algebra omitted from Section 4.3. Let $\mathbf{V}_{\text{tra}}$ be the retained orthonormal POD basis of Eq. (45), with $n_{\text{tra}} = n_{\text{tot}}$ columns, let $\mathbf{S}_{\text{snap}}$ collect the centered displacement snapshots, and collect the corresponding engineering-strain inputs in $\mathbf{M} = [\mathbf{e}^1, \dots, \mathbf{e}^{N_s}]$. Their POD coefficients are

$$\mathbf{Q}_{\text{POD}} = \mathbf{V}_{\text{tra}}^T \mathbf{S}_{\text{snap}}. \quad (88)$$

The input-informed construction first identifies the least-squares map from these coefficients to strain,

$$\mathbf{T}_m = \mathbf{M} \mathbf{Q}_{\text{POD}}^\dagger \in \arg \min_{\mathbf{H}} \|\mathbf{H} \mathbf{Q}_{\text{POD}} - \mathbf{M}\|_F^2, \quad (89)$$

where $\dagger$ denotes the Moore–Penrose pseudoinverse. Assuming rank three, the compact factorization

$$\mathbf{V}_{\text{tra}} \mathbf{T}_m^T = \mathbf{V} \Sigma_m \mathbf{Z}_m^T, \quad \mathbf{A}_m = \Sigma_m^{-1} \mathbf{Z}_m^T \quad (90)$$

defines the three-column orthonormal primary block $\mathbf{V}$ and the coordinate transformation $\mathbf{A}_m$, with $\mathbf{q} = \mathbf{A}_m \boldsymbol{\xi}$ and $\boldsymbol{\xi}^s = \mathbf{T}_m \mathbf{Q}_{\text{POD}}^s$. The secondary block $\bar{\mathbf{V}}$ is an orthonormal complement within the retained POD span, ordered by snapshot energy, so that $\mathbf{V}_{\text{tot}} = [\mathbf{V}, \bar{\mathbf{V}}]$ spans the same subspace as $\mathbf{V}_{\text{tra}}$ and differs from it only by an orthogonal change of basis. The two blocks give the deployed decoder of Eq. (52).

The displacement error separates the two approximation mechanisms. For the projected coordinates $(\mathbf{q}^s, \bar{\mathbf{q}}^s)$,

$$\|\mathbf{u}^s - \tilde{\mathbf{u}}(\mathbf{q}^s)\|_2^2 = \|\bar{\mathbf{q}}^s - \mathcal{N}(\mathbf{q}^s)\|_2^2 + \|(I - \mathbf{V}_{\text{tot}} \mathbf{V}_{\text{tot}}^T)(\mathbf{u}^s - \mathbf{u}_{\text{ref}})\|_2^2. \quad (91)$$

The first term is the closure error within the retained span, and the second is the POD truncation error, which no increase in network capacity can recover.

Finally, let $\mathbf{B}_\xi = \partial_{\boldsymbol{\xi}} \tilde{\mathbf{u}}$, $\mathbf{R}$ be the FOM residual, and $\mathbf{K} = \partial_{\mathbf{u}} \mathbf{R}$. Differentiating the projected residual $\mathbf{r} = \mathbf{B}_\xi^T \mathbf{R}$ gives

$$\frac{\partial \mathbf{r}}{\partial \boldsymbol{\xi}} = \mathbf{B}_\xi^T \mathbf{K} \mathbf{B}_\xi + \sum_{j=1}^N R_j \frac{\partial^2 \tilde{u}_j}{\partial \boldsymbol{\xi} \partial \boldsymbol{\xi}}. \quad (92)$$

The second term is the curvature contribution of the decoder manifold. It need not vanish at reduced equilibrium because $\mathbf{B}_\xi^T \mathbf{R} = \mathbf{0}$ does not imply $\mathbf{R} = \mathbf{0}$.

E Elementwise cubature conventions

This appendix records the implementation conventions omitted from Section 4. The general ECM construction and MAW–ECM pruning algorithm follow Hernández et al. [34, 39], and the expressions below specify how the deployed residual and stress rules are assembled.

For the fixed rule, let $\mathbf{c}_e^s \in \mathbb{R}^d$ contain the contribution of element e at training state $s = 1, \dots, N_c$. Residual cubature uses $\mathbf{c}_e^s = \mathbf{h}_e^s$. Stress cubature uses the four components of

$$\mathbf{p}_e^s = \frac{1}{|Y|} \sum_{g \in e} \omega_g \mathbf{P}_{\mu, g}^s, \quad \mathbf{c}_e^s = \text{vec } \mathbf{p}_e^s = \begin{bmatrix} p_{e,11}^s & p_{e,12}^s & p_{e,21}^s & p_{e,22}^s \end{bmatrix}^T. \quad (93)$$

One element occupies each row of the sampled integrand matrix,

$$\mathbf{M}_{\text{int}} = \begin{bmatrix} (\mathbf{c}_1^1)^T & \dots & (\mathbf{c}_1^{N_c})^T \\ \vdots & & \vdots \\ (\mathbf{c}_{N_{\text{el}}}^1)^T & \dots & (\mathbf{c}_{N_{\text{el}}}^{N_c})^T \end{bmatrix} \in \mathbb{R}^{N_{\text{el}} \times (dN_c)}. \quad (94)$$

A truncated SVD gives $\mathbf{M}_{\text{int}} \simeq \mathbf{U}_{\text{cub}} \mathbf{\Sigma}_{\text{cub}} \mathbf{V}_{\text{cub}}^T$. To enforce the multiplier sum without duplicating a constant component already represented by the retained integration modes, the fitting system uses

$$\mathbf{p}_{\perp} = (\mathbf{I} - \mathbf{U}_{\text{cub}} \mathbf{U}_{\text{cub}}^T) \mathbf{1}, \quad \mathbf{G}_{\text{cub}} = \begin{bmatrix} \mathbf{U}_{\text{cub}}^T \\ \mathbf{p}_{\perp}^T / \|\mathbf{p}_{\perp}\|_2 \end{bmatrix}, \quad \mathbf{b}_{\text{cub}} = \mathbf{G}_{\text{cub}} \mathbf{1}. \quad (95)$$

If the constant already lies in the retained span, the last row is redundant. Otherwise its exact reproduction gives $\sum_e w_e = N_{\text{el}}$. The reference quadrature measures are already contained in $\mathbf{c}_e^s$, so the unknowns are element multipliers.

After MAW–ECM pruning selects a support $\mathcal{Z} = \{e_1, \dots, e_{|\mathcal{Z}|}\}$, neural fitting returns to the uncompressed contributions at each decoded state and targets the full-mesh sum,

$$\mathbf{A}^s = \begin{bmatrix} \mathbf{c}_{e_1}^s & \dots & \mathbf{c}_{e_{|\mathcal{Z}|}}^s \\ 1 & \dots & 1 \end{bmatrix}, \quad \mathbf{b}^s = \begin{bmatrix} \sum_{e \in \mathcal{E}} \mathbf{c}_e^s \\ N_{\text{el}} \end{bmatrix}. \quad (96)$$

Residual fitting uses $d = n$ and $\mathcal{Z}_r$, and stress fitting uses $d = 4$ and $\mathcal{Z}_s$. With the scaled-softmax map of Eq. (58), the residual network minimizes

$$\mathcal{L}_{\text{cub}}(\boldsymbol{\theta}_r) = \frac{1}{|\mathcal{I}_{\text{fit}}|} \sum_{s \in \mathcal{I}_{\text{fit}}} \frac{\|\mathbf{A}^s \mathbf{w}(\boldsymbol{\xi}^s) - \mathbf{b}^s\|_2^2}{\max(\|\mathbf{b}^s\|_2^2, \delta_{\text{floor}})}, \quad (97)$$

and the stress network uses the analogous loss with $\mathbf{v}$. A preliminary fit to the sampled MAW–ECM weights initializes each network, but the final objective is integration accuracy, because a reproducing weight vector is not unique.

For the tangent, both reduced implementations use Eq. (61) but evaluate its partial derivatives differently. For the affine HPROM, the assembled reduced stiffness and the analytic lifting derivative provide $d\mathbf{q}_{\text{tra}}^*/d\mathbf{e}$ through three linear right-hand sides, and central differences are applied only to the stress along those linearized equilibrium directions. For HPROM–ANN, central differences at fixed $\mathbf{q}$ or fixed $\mathbf{e}$ evaluate all four partial-derivative blocks in Eq. (61), including decoder curvature and weight variation. The resulting small linear system does not require a new nonlinear microscopic solve for every perturbed strain. Numerical differentiation is an implementation choice, and analytic or automatic differentiation would give the same formal tangent up to their respective numerical tolerances.

F Capacity and feature-count controls

The comparisons of Section 5 pair compact constrained cores with a larger Unconstrained network. To separate the effect of the constraints from that of capacity, both benchmarks complete a two-by-two design. The constrained cores are widened to about 25 000 parameters, comparable with the Unconstrained energy, and an Unconstrained energy with 981 parameters matches the compact ICNN. Every new fit follows the protocol of Section 5.1.1 with the same three seeds, and its test states are opened only once, after training has ended.

Table 6 reports the MC-RVE cells as medians over three seeds, as in Table 3. Widening changes the ICNN error from 0.0445% to 0.0398%, while the Unconstrained energy improves from 0.0126% to 0.0073%.

At matched capacity the constraints therefore cost a factor of 3.2 to 5.5 in stress error. The wide ICKAN is less accurate than the compact one because two of its three seeds met the validation-plateau criterion early, so its row reflects the common stopping rule rather than capacity.

Table 7 reports the SC-RVE cells for the validation-selected seed, as in Table 4, together with the coupon of Section 5.3.5. Widening changes the constrained test errors by less than 3%, and the Unconstrained energy with 981 parameters remains about 75 to 80 times more accurate than either wide core. In the coupon, the corresponding factor between matched cells lies between about 115 and 200. Thirty-two learned features, trained with the same protocol, lower the test errors to 0.537% and 0.618% and the coupon errors to 0.343% and 0.347%. They also raise the probe errors to 35.6% and 16.9% (Section 5.3.2), and the sufficient bound of Appendix C fails for both saved parameter sets. The six-feature models therefore remain the deployed choice.

Table 6: MC-RVE capacity controls: independent-test errors, in percent.

Model	Parameters	Stress [%]	Energy [%]	Tangent [%]
ICNN, $m = 6$	975	0.0445	0.0079	0.3851
ICNN, wide	26 919	0.0398	0.0069	0.2962
ICKAN, $m = 6$	1 659	0.0406	0.0066	0.3622
ICKAN, wide*	24 847	0.0744	0.0104	1.8311
Unconstrained, small	981	0.0126	0.0019	0.1579
Unconstrained	25 473	0.0073	0.0011	0.0857

Relative error norm over the 512 test states, median over three seeds. *Adam stopped at the validation plateau after 28 540 and 33 330 steps for two of the three seeds, against 114 910 for the third.

Table 7: SC-RVE capacity and feature-count controls, in percent.

Model	Parameters	e_S^{test} [%]	e_u [%]
ICNN, $m = 6$	975	0.6214	0.4066
ICNN, wide	26 919	0.6066	0.4162
ICNN, $m = 32$	2 379	0.5370	0.3433
ICKAN, $m = 6$	1 659	0.6359	0.4197
ICKAN, wide	24 847	0.6540	0.4322
ICKAN, $m = 32$	3 999	0.6177	0.3469
Unconstrained, small	981	0.0081	0.0035
Unconstrained	25 473	0.0052	0.0022

Validation-selected seed of each cell. Test: 400 states; e_u : final nodal displacement error of the coupon of Section 5.3.5.

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